INVERSE TRIGONOMETRIC FUNCTIONS

SYNOPSIS

- **Inverse of a function :**

$f: A \to B$ is bijective $\Leftrightarrow f^{-1}: B \to A$ exists and it is also bijective. All trigonometric functions are not bijective functions. By restricting the domains of the functions,we make them bijective

- The function $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right] \to [-1,1]$ defined by $f(x)=\sin x$ is a bijection.

Then $f^{-1}:[-1,1] \to \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ is also a bijection. This function is called **inverse sine** function and it is denoted by **Arc sine** x or $Sin^{-1}x$

- The function $f:[0,\pi] \to [-1,1]$ defined by $f(x)=\cos x$ is a bijection.

Then $f^{-1}:[-1,1] \to [0,\pi]$ is also a bijection. This function is called **inverse cosine** function and it is denoted by **Arc cos** x or $Cos^{-1}x$

- The function $f:\left(-\frac{\pi}{2},\frac{\pi}{2}\right) \to R$ defined by $f(x)=\tan x$ is a bijection.

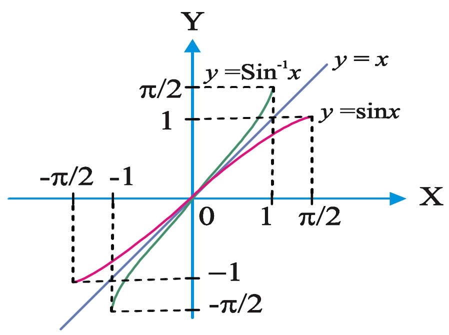
Then $f^{-1}: \mathrm{R} \to \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ is also a bijection. This function is called **inverse tangent** function and it is denoted by **Arc tan** x or $Tan^{-1}x$

- **Domains and Ranges of Inverse trigonometric functions:**

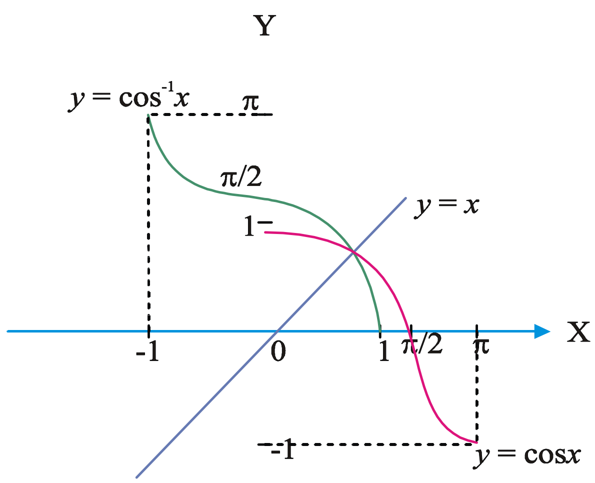
Function	Domain	Range
$Sin^{-1}x$	$[-1,1]$	$\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$
$Cos^{-1}x$	$[-1,1]$	$[0,\pi]$
$Tan^{-1}x$	R	$\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$
$Cot^{-1}x$	R	$(0,\pi)$
$Sec^{-1}x$	$(-\infty,-1]\cup[1,\infty)$	$\left[0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right]$
$Cosec^{-1}x$	$(-\infty,-1]\cup[1,\infty)$	$\left[\frac{-\pi}{2},0\right)\cup\left(0,\frac{\pi}{2}\right]$

- **Graphs of inverse circular functions:**

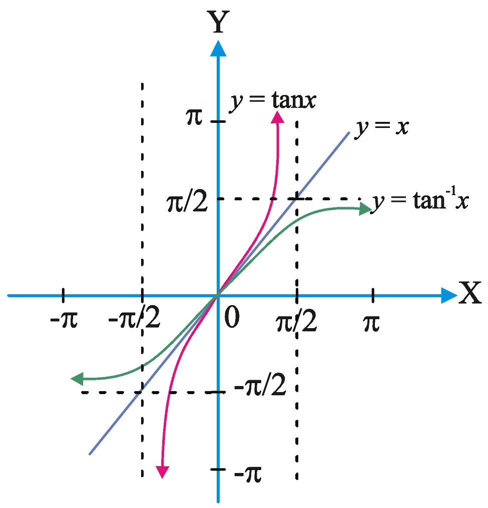
1. $y = Sin^{-1}x, |x| \le 1, y \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$



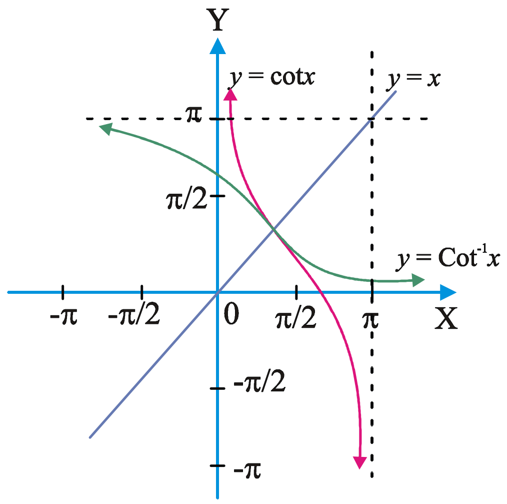
2. $y = Cos^{-1}x, |x| \le 1, y \in [0,\pi]$



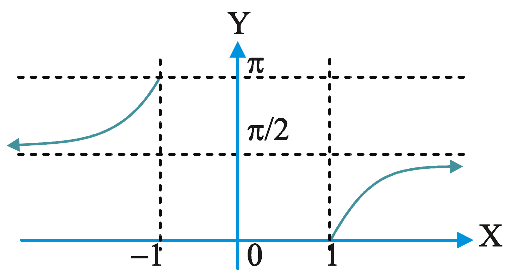
3. $y = \mathrm{Tan}^{-1} x,\ x \in R,\ y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



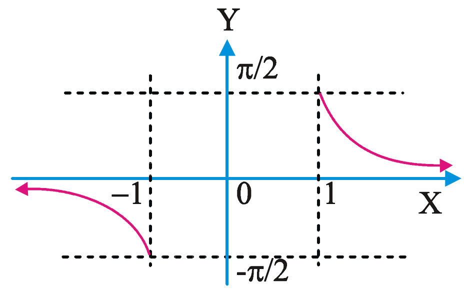
4. $y = \mathrm{Cot}^{-1} x,\ x \in R, \quad y \in (0, \pi)$



5. $y = \mathrm{Sec}^{-1} x,\ |x| \geq 1,\ \ y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$



6. $y = \mathrm{Cosec}^{-1}x,\ |x| \geq 1, \quad y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$



Properties of inverse trigonometric functions:

➢ i) $Sin^{-1}x = Cosec^{-1}\frac{1}{x}, \quad \forall x \in [-1,1], x \neq 0$

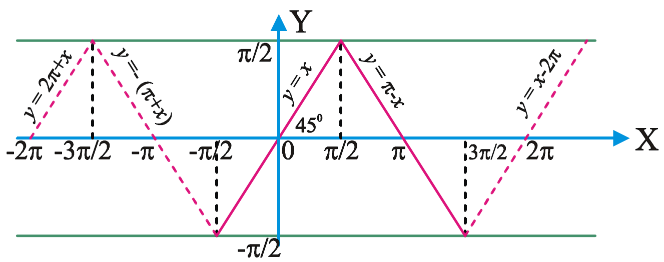
ii) $Cos^{-1}x = Sec^{-1}\frac{1}{x}, \quad \forall x \in [-1,1], x \neq 0$

iii) $Tan^{-1}x = \begin{cases} Cot^{-1}\frac{1}{x}, & \forall x > 0 \\ -\pi + Cot^{-1}\frac{1}{x}, & \forall x < 0 \end{cases}$

➢ Some useful periodic graphs:

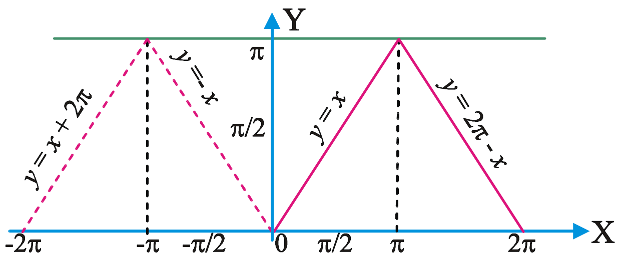
1. $$y = Sin^{-1}(\sin x) = \begin{cases} -\pi - x, & -\frac{3\pi}{2} \leq x \leq -\frac{\pi}{2} \\ x, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} \leq x \leq \frac{3\pi}{2} \\ -2\pi + x, & \frac{3\pi}{2} \leq x \leq \frac{5\pi}{2} \end{cases}$$

y is Periodic with period 2π and $y \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$



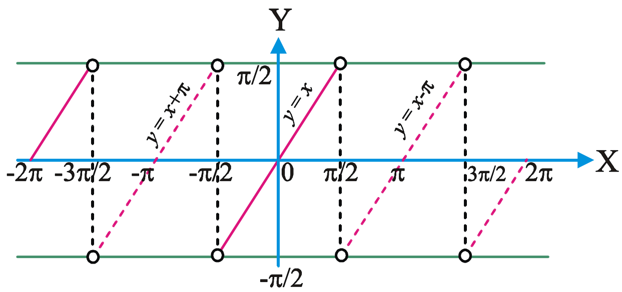
2. $y = Cos^{-1}(\cos x) = \begin{cases} -x, & -\pi \le x \le 0 \\ x, & 0 \le x \le \pi \\ 2\pi - x, & \pi \le x \le 2\pi \\ -2\pi + x, & 2\pi \le x \le 3\pi \end{cases}$

y is periodic with period 2π and $y \in [0, \pi]$



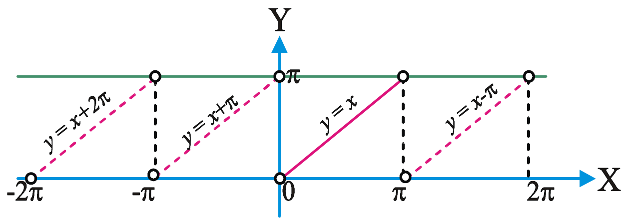
3. $y = Tan^{-1}(\tan x) = \begin{cases} x + \pi, & -\frac{3\pi}{2} < x < -\frac{\pi}{2} \\ x, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -\pi + x, & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ x - 2\pi, & \frac{3\pi}{2} < x < \frac{5\pi}{2} \end{cases}$

y is periodic with period π and $y \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$



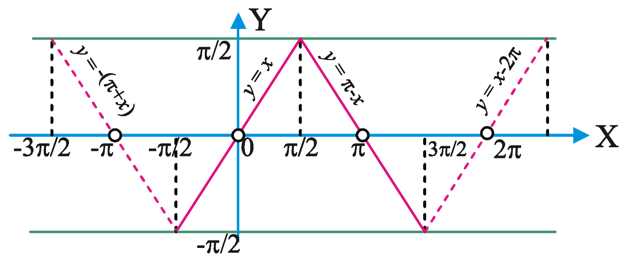
4. $y = Cot^{-1}(Cot\, x) = x, \quad x \in (0, \pi)$ and so on.

y is periodic with period π and $y \in (0, \pi)$



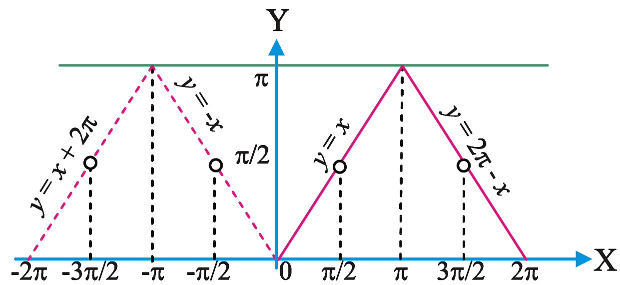
5. $y = Cosec^{-1}(cosec\, x) = \begin{cases} x, & x \in \left[\frac{-\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \\ \pi - x, & x \in \left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right] \\ \text{and so on} & \end{cases}$

y is periodic with period 2π and $y \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$



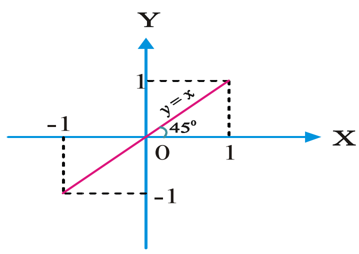
6. $y = Sec^{-1}(\sec x) = \begin{cases} x, & x \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right] \\ 2\pi - x, & x \in \left[\pi, \frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right] \\ \text{and so on} & \end{cases}$

y is periodic with period 2π and $y \in [0, \pi]$

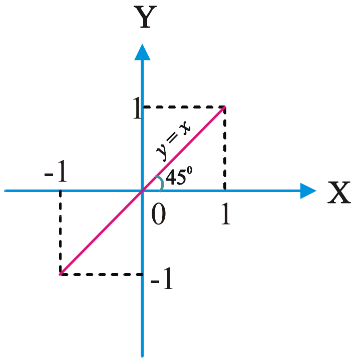


➢ Some useful non-periodic graphs:

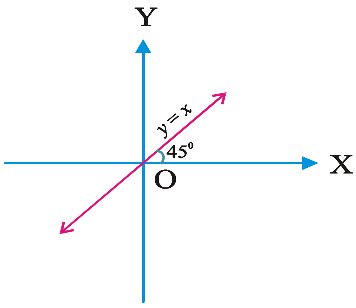
1. $y = Sin(Sin^{-1}x) = x,\ x \in [-1, 1],\ \ y \in [-1, 1]$



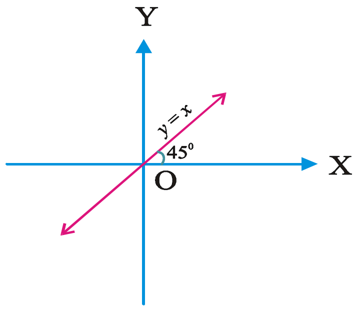
2. $y = Cos(Cos^{-1}x) = x,\ x \in [-1, 1],\ y \in [-1, 1]$



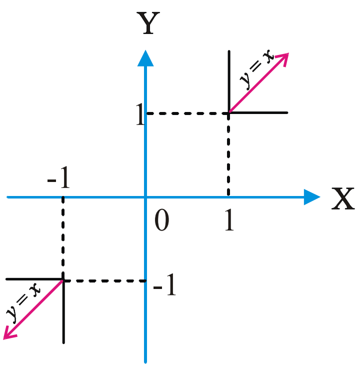
3. $y = Tan\ (Tan^{-1}x) = x,\ x \in R,\ \ y \in R$



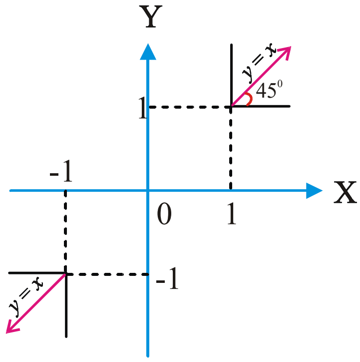
4. $y = Cot\ (Cot^{-1}x) = x,\ x \in R,\ y \in R$



5. $y = \operatorname{cosec}\left(\operatorname{cosec}^{-1}x\right) = x,\ |\,x\,| \geq 1,\ |\,y\,| \geq 1,$



6. $y = Sec(Sec^{-1}x) = x,\ |\,x\,| \geq 1,\ |\,y\,| \geq 1$



➢ **Important Results:**

i) $Sin^{-1}(-x) = -Sin^{-1}x,\ \forall\ x \in [-1,1]$

ii) $Cos^{-1}(-x) = \pi - Cos^{-1}x,\ \forall\ x \in [-1,1]$

iii) $Tan^{-1}(-x) = -Tan^{-1}x,\ \forall\ x \in R$

iv) $Cosec^{-1}(-x) = -Cosec^{-1}x,\ \forall\ x \in R-(-1,1)$

v) $Cot^{-1}(-x) = \pi - Cot^{-1}x,\ \forall\ x \in R$

vi) $Sec^{-1}(-x) = \pi - Sec^{-1}x,\ \forall\ x \in (-\infty,-1] \cup [1,\infty)$

W.E-1: $Cos^{-1}\left(\frac{-1}{2}\right) - 2Sin^{-1}\left(\frac{1}{2}\right) + 3Cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$

$-4Tan^{-1}(-1) =$ ***(EAM-2009)***

Sol: $Cos^{-1}\left(\frac{-1}{2}\right) - 2Sin^{-1}\left(\frac{1}{2}\right) + 3Cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$

$-4Tan^{-1}(-1) = \pi - Cos^{-1}\left(\frac{1}{2}\right) - 2\left(\frac{\pi}{6}\right)$

$+3\left(\pi - Cos^{-1}\left(\frac{1}{\sqrt{2}}\right)\right) + 4Tan^{-1}(1)$

$= \pi - \frac{\pi}{3} - \frac{\pi}{3} + 3\left(\pi - \frac{\pi}{4}\right) + 4\left(\frac{\pi}{4}\right)$

$= \frac{\pi}{3} + 3\left(\frac{3\pi}{4}\right) + \pi = \frac{43\pi}{12}$

➢ i) $\theta \in [0, \pi]$ then $Sin^{-1}(\cos\theta) = \frac{\pi}{2} - \theta$

ii) $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ then $Cos^{-1}(\sin\theta) = \frac{\pi}{2} - \theta$

iii) $\theta \in (0,\pi)$ then $Tan^{-1}(\cot\theta) = \frac{\pi}{2} - \theta$

iv) $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then $Cot^{-1}(\tan\theta) = \frac{\pi}{2} - \theta$

v) $\theta \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ then

$$Sec^{-1}(\operatorname{cosec}\theta) = \frac{\pi}{2} - \theta$$

vi) $\theta \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ then

$$Cosec^{-1}(\sec\theta) = \frac{\pi}{2} - \theta$$

➤ i) $$Sin^{-1}x = \begin{cases} Cos^{-1}\sqrt{1-x^2} & if\ \ 0 \le x \le 1 \\ -Cos^{-1}\sqrt{1-x^2} & if\ \ -1 \le x < 0 \\ Tan^{-1}\dfrac{x}{\sqrt{1-x^2}} & if\ \ x \in (-1,1) \end{cases}$$

ii) $$Cos^{-1}x = \begin{cases} Sin^{-1}\sqrt{1-x^2} & if\ \ x \in [0,1] \\ \pi - Sin^{-1}\sqrt{1-x^2} & if\ \ x \in [-1,0) \end{cases}$$

iii) $$Tan^{-1}x = \begin{cases} Sin^{-1}\dfrac{x}{\sqrt{1+x^2}} & for\ \ x > 0 \\ Cos^{-1}\dfrac{1}{\sqrt{1+x^2}} & for\ \ x > 0 \end{cases}$$

W.E-2: The value of x, where $x > 0$ and

$Tan\left(Sec^{-1}\frac{1}{x}\right) = Sin\left(Tan^{-1}2\right)$ ***is (EAM-2007)***

Sol: $Tan\left(Sec^{-1}\frac{1}{x}\right) = Sin\left(Tan^{-1}2\right)$

$$Tan\left(Tan^{-1}\frac{\sqrt{1-x^2}}{x}\right) = Sin\left(Sin^{-1}\frac{2}{\sqrt{1+2^2}}\right)$$

$$\left(\because Tan^{-1}x = Sin^{-1}\frac{x}{\sqrt{1+x^2}}\right)$$

$$\frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}} \Rightarrow 5(1-x^2) = 4x^2$$

$$\Rightarrow x^2 = \frac{5}{9} \Rightarrow x = \frac{\sqrt{5}}{3}$$

➤ i) $Sin^{-1}x + Cos^{-1}x = \frac{\pi}{2}, \forall\ x \in [-1,1]$

ii) $Tan^{-1}x + Cot^{-1}x = \pi/2, \forall\ x \in R$

iii) $Sec^{-1}x + Cosec^{-1}x = \frac{\pi}{2}, \forall\ x \in (-\infty,-1] \cup [1,\infty)$

$$Sin^{-1}x + Sin^{-1}y = \begin{cases} Sin^{-1}\left\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\right\} \\ 0 \le x, y, x^2 + y^2 \le 1 \text{ or} \\ -1 \le x, y \le 1, xy < 0 \ and\ x^2 + y^2 > 1 \\ \pi - Sin^{-1}\left\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\right\} \\ if\ 0 < x, y \le 1 \text{ and } x^2 + y^2 > 1 \\ -\pi - Sin^{-1}\left\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\right\} \\ \text{if } -1 \le x, y < 0 \text{ and } x^2 + y^2 > 1 \end{cases}$$

$$Sin^{-1}x - Sin^{-1}y = \begin{cases} Sin^{-1}\left\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\right\} \\ 0 \le x, y < 1 \text{ and } x^2 + y^2 > 1 \text{ or} \\ -1 \le x, y \le 1, xy < 0 \text{ and } x^2 + y^2 \le 1 \\ \pi - Sin^{-1}\left\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\right\} \\ if\ 0 < x \le 1, -1 \le y < 1 \text{ and } x^2 + y^2 > 1 \\ -\pi - Sin^{-1}\left\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\right\} \\ \text{if } 0 < y \le 1, -1 \le x < 0 \text{ and } x^2 + y^2 > 1 \end{cases}$$

$$Cos^{-1}x + Cos^{-1}y = \begin{cases} Cos^{-1}\left\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\right\} \\ if\ -1 \le x, y \le 1 \text{ and } x + y \ge 0 \\ 2\pi - Cos^{-1}\left\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\right\} \\ if\ -1 \le x, y \le 1 \text{ and } x + y < 0 \end{cases}$$

$$Cos^{-1}x - Cos^{-1}y = \begin{cases} Cos^{-1}\left\{xy+\sqrt{1-x^2}\sqrt{1-y^2}\right\} \\ if\ -1\le x, y\le 1, x\le y \\ -Cos^{-1}\left\{xy+\sqrt{1-x^2}\sqrt{1-y^2}\right\} \\ if\ -1\le y\le 0, 0< x\le 1 \text{ and } x\ge y \end{cases}$$

W.E-3: If $\frac{1}{2}\le x\le 1$ ***then***

$Cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)+Cos^{-1}x$ ***is equal to***

(EAM-2012)

Sol: $Cos^{-1}x + Cos^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)$

$= Cos^{-1}x + Cos^{-1}\left(x.\frac{1}{2}+\frac{\sqrt{3}}{2}\sqrt{1-x^2}\right)$

$= Cos^{-1}x + Cos^{-1}\left(\frac{1}{2}\right) - Cos^{-1}x = Cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$

$$Tan^{-1}x + Tan^{-1}y = \begin{cases} Tan^{-1}\left(\frac{x+y}{1-xy}\right) if\ x>0, y>0, xy<1 \\ \pi + Tan^{-1}\left(\frac{x+y}{1-xy}\right) if\ x>0, y>0, xy>1 \\ -\pi + Tan^{-1}\left(\frac{x+y}{1-xy}\right) if\ x<0, y<0, xy>1 \\ \frac{\pi}{2}, if\ xy=1 \end{cases}$$

$$Tan^{-1}x - Tan^{-1}y = \begin{cases} Tan^{-1}\left(\frac{x-y}{1+xy}\right) for\ xy>-1 \\ \pi + Tan^{-1}\left(\frac{x-y}{1+xy}\right) if\ x>0, y<0 \text{ and } xy<-1 \\ -\pi + Tan^{-1}\left(\frac{x-y}{1+xy}\right) if\ x<0, y>0 \text{ and } xy<-1 \end{cases}$$

➢ (i) If x, y, z have same sign and $xy+yz+zx<1$ then

$$Tan^{-1}x + Tan^{-1}y + Tan^{-1}z = Tan^{-1}\left(\frac{x+y+z-xyz}{1-xy-yz-zx}\right)$$

(ii) $\mathrm{Tan}^{-1}x_1 + \mathrm{Tan}^{-1}x_2 + ...\mathrm{Tan}^{-1}x_n =$

$$\mathrm{Tan}^{-1}\left(\frac{S_1 - S_3 + S_5 - ....}{1 - S_2 + S_4 - S_6....}\right)$$

where S_1 = sum of values , S_2 = sum of product of taken two elements at a time and so on..,

S_n = product of values.

W.E-4: The value of $Cot\left(Cosec^{-1}\frac{5}{3}+Tan^{-1}\frac{2}{3}\right)=$

(AIE-2008)

Sol: $Cot\left(Cosec^{-1}\frac{5}{3}+Tan^{-1}\frac{2}{3}\right)=Cot\left(Sin^{-1}\frac{3}{5}+Tan^{-1}\frac{2}{3}\right)$

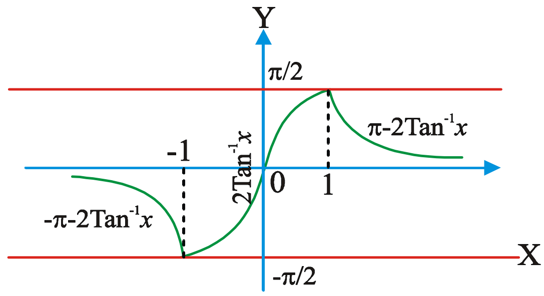
$=Cot\left(Tan^{-1}\frac{3}{4}+Tan^{-1}\frac{2}{3}\right)\ \left[\because Sin^{-1}x = Tan^{-1}\frac{x}{\sqrt{1-x^2}}\right]$

$=Cot\left[Tan^{-1}\left(\frac{\frac{3}{4}+\frac{2}{3}}{1-\left(\frac{3}{4}\right)\left(\frac{2}{3}\right)}\right)\right]=Cot\left[Tan^{-1}\left(\frac{\frac{17}{12}}{\frac{1}{2}}\right)\right]$

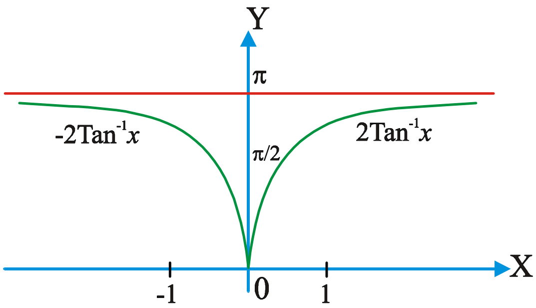
$=\cot\cot^{-1}\left(\frac{17}{6}\right)=\cot\tan^{-1}\left(\frac{17}{6}\right)=\frac{6}{17}$

➢ **Transformation of Inverse functions by elementry substitution and their graphs:**

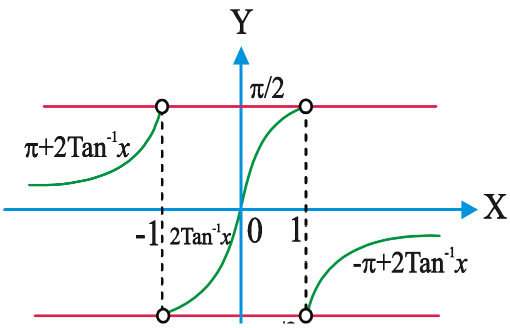
$$i)\ \mathrm{Sin}^{-1}\frac{2x}{1+x^2} = \begin{cases} -\pi - 2\,\mathrm{Tan}^{-1}x & x\le -1 \\ 2\,\mathrm{Tan}^{-1}x & -1\le x\le 1 \\ \pi - 2\,\mathrm{Tan}^{-1}x & x\ge 1 \end{cases}$$



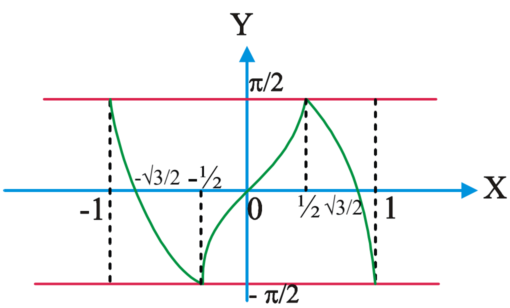
ii) $\mathrm{Cos}^{-1}\dfrac{1-x^2}{1+x^2}=\begin{cases}2\,\mathrm{Tan}^{-1}x & x\geq 0\\ -2\,\mathrm{Tan}^{-1}x & x<0\end{cases}$



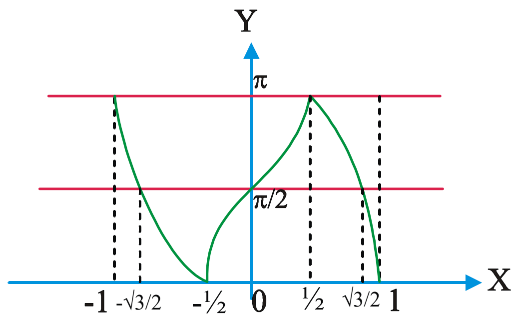
iii) $\mathrm{Tan}^{-1}\dfrac{2x}{1-x^2}=\begin{cases}\pi+2\,\mathrm{Tan}^{-1}x & x<-1\\ 2\,\mathrm{Tan}^{-1}x & -1<x<1\\ -\pi+2\,\mathrm{Tan}^{-1}x & x>1\end{cases}$



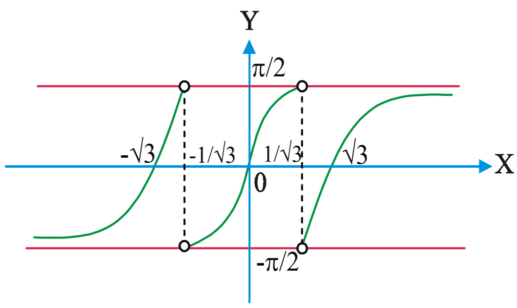
➢i) $Sin^{-1}\left(3x-4x^3\right)=\begin{cases}-\left(\pi+3Sin^{-1}x\right) & if\ -1\leq x\leq -1/2\\ 3Sin^{-1}x & if\ -1/2\leq x\leq 1/2\\ \pi-3Sin^{-1}x & if\ 1/2\leq x\leq 1\end{cases}$



ii) $Cos^{-1}\left(4x^3-3x\right)=\begin{cases}3\cos^{-1}x-2\pi & if\ -1\leq x\leq -1/2\\ 2\pi-3Cos^{-1}x & if\ -1/2\leq x\leq 1/2;\\ 3Cos^{-1}x & if\ 1/2\leq x\leq 1\end{cases}$



iii) $\mathrm{Tan}^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right)=\begin{cases}3\,\mathrm{Tan}^{-1}x & if\ -\dfrac{1}{\sqrt{3}}<x<\dfrac{1}{\sqrt{3}}\\ \pi+3\,\mathrm{Tan}^{-1}x & if\ x>\dfrac{1}{\sqrt{3}}\\ -\pi+3\,\mathrm{Tan}^{-1}x & if\ x<-\dfrac{1}{\sqrt{3}}\end{cases}$



➢i) $2Sin^{-1}x=\begin{cases}-\pi-Sin^{-1}\left(2x\sqrt{1-x^2}\right) & if\ -1\leq x\leq -\dfrac{1}{\sqrt{2}}\\ Sin^{-1}\left(2x\sqrt{1-x^2}\right) & if\ -\dfrac{1}{\sqrt{2}}\leq x\leq \dfrac{1}{\sqrt{2}}\\ \pi-Sin^{-1}\left(2x\sqrt{1-x^2}\right) & if\ \dfrac{1}{\sqrt{2}}\leq x\leq 1\end{cases}$

ii) $2Cos^{-1}x=\begin{cases}Cos^{-1}\left(2x^2-1\right) & if\ 0\leq x\leq 1\\ 2\pi-Cos^{-1}\left(2x^2-1\right) & if\ -1\leq x\leq 0\end{cases}$

iii) $2Tan^{-1}x=\begin{cases}Tan^{-1}\left(\dfrac{2x}{1-x^2}\right) & if\ -1<x<1\\ \pi+Tan^{-1}\left(\dfrac{2x}{1-x^2}\right) & if\ x>1\\ -\pi+Tan^{-1}\left(\dfrac{2x}{1-x^2}\right) & if\ x<-1\end{cases}$

iv) $$2Tan^{-1}x = \begin{cases} -\pi - Sin^{-1}\left(\frac{2x}{1+x^2}\right) & if\ x<-1 \\ Sin^{-1}\left(\frac{2x}{1+x^2}\right) & if\ -1\le x\le 1 \\ \pi - Sin^{-1}\left(\frac{2x}{1+x^2}\right) & if\ x>1 \end{cases}$$

v) $$2Tan^{-1}x = \begin{cases} -Cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & if\ -\infty < x \le 0 \\ Cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & if\ 0\le x<\infty \end{cases}$$

W.E-5: $Sin^{-1}\frac{4}{5}+2Tan^{-1}\frac{1}{3}=$ **(EAM-05)**

Sol: $Sin^{-1}\frac{4}{5}+2Tan^{-1}\frac{1}{3}$

$$=Sin^{-1}\frac{4}{5}+Tan^{-1}\left(\frac{2\left(\frac{1}{3}\right)}{1-\left(\frac{1}{3}\right)^2}\right)\left[\because 2Tan^{-1}x = Tan^{-1}\left(\frac{2x}{1-x^2}\right)\right]$$

$$=Sin^{-1}\frac{4}{5}+Tan^{-1}\frac{3}{4}$$

$$=Sin^{-1}\frac{4}{5}+Cos^{-1}\frac{4}{5}\left[\because Tan^{-1}x=Cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right]$$

$$=\frac{\pi}{2}\quad\left[\because Sin^{-1}x+Cos^{-1}x=\frac{\pi}{2}\right]$$

➢ Some important facts:

1) $Tan^{-1}x+Tan^{-1}y+Tan^{-1}z=\frac{\pi}{2}$, $if\ xy+yz+zx=1$

2) $Tan^{-1}x+Tan^{-1}y+Tan^{-1}z=\pi$, $if\ x+y+z=xyz$ **(EAM-2014)**

3) $Tan^{-1}\frac{a}{x}+Tan^{-1}\frac{b}{x}=\frac{\pi}{2}$,then $x=\sqrt{ab}$

4) $Sin^{-1}\frac{a}{x}+Sin^{-1}\frac{b}{x}=\frac{\pi}{2}$, then $x=\sqrt{a^2+b^2}$

5) $Tan^{-1}\left(\frac{p}{q}\right)+Tan^{-1}\left(\frac{q-p}{q+p}\right)=\frac{\pi}{4}$

6) $Tan^{-1}\left(\frac{1+x}{1-x}\right)=\frac{\pi}{4}+\tan^{-1}x\quad if\ x<1$

7) $Tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{\pi}{4}-\tan^{-1}x\quad if\ x>-1$

8) If $Tan^{-1}x+Tan^{-1}y=\frac{\pi}{2}$ then xy=1.

9) $Cot^{-1}x+Cot^{-1}y=\frac{\pi}{2}$ then xy=1

10) If $\mathrm{Cos}^{-1}x+\mathrm{Cos}^{-1}y+\mathrm{Cos}^{-1}z=3\pi$ then $xy+yx+zx=3$

11) If $\mathrm{Sin}^{-1}x+\mathrm{Sin}^{-1}y+\mathrm{Sin}^{-1}z=\frac{3\pi}{2}$ then $xy+yx+zx=3$

12) If $\mathrm{Sin}^{-1}x+\mathrm{Sin}^{-1}y=\theta$ then $\mathrm{Cos}^{-1}x+\mathrm{Cos}^{-1}y=\pi-\theta$

13) If $\mathrm{Cos}^{-1}x+\mathrm{Cos}^{-1}y=\theta$ then $\mathrm{Sin}^{-1}x+\mathrm{Sin}^{-1}y=\pi-\theta$

14) If $a\mathrm{Sin}^{-1}x-b\mathrm{Cos}^{-1}x=c$ then

$$a\mathrm{Sin}^{-1}x+b\mathrm{Cos}^{-1}x=\frac{\pi ab+c(a-b)}{a+b}$$

15) If $Cos^{-1}\frac{x}{a}+Cos^{-1}\frac{y}{b}=\theta$ then

$$\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\theta+\frac{y^2}{b^2}=Sin^2\theta$$

16) If $Sin^{-1}\frac{x}{a}+Sin^{-1}\frac{y}{b}=\theta$ then

$$\frac{x^2}{a^2}+\frac{2xy}{ab}\cos\theta+\frac{y^2}{b^2}=Sin^2\theta$$

17) $Tan^{-1}\left(\frac{1}{1+x(x+1)}\right)+Tan^{-1}\left(\frac{1}{1+(x+1)(x+2)}\right)+....+$

$$Tan^{-1}\left(\frac{1}{1+(x+n-1)(x+n)}\right)=Tan^{-1}(x+n)-Tan^{-1}x, n\in N$$

➢ If an expression contains

i) $\sqrt{a^2-x^2}$, *put* $x=a\sin\theta, \theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

ii) $\sqrt{a^2+x^2}$, *put* $x=a\tan\theta, \theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

iii) $\sqrt{x^2-a^2}$, *put* $x=a\sec\theta, \theta\in(0,\pi)$

➢ **Range of some special inverse Trigonometric Functions :**

i) $\frac{\pi^3}{32}\le(\mathrm{Sin}^{-1}x)^3+(\mathrm{Cos}^{-1}x)^3\le\frac{7\pi^3}{8}$

ii) $\frac{\pi^2}{8}\le(\mathrm{Sin}^{-1}x)^2+(\mathrm{Cos}^{-1}x)^2\le\frac{5\pi^2}{4}$

iii) $-\frac{\pi^2}{4}\le(\mathrm{Cos}^{-1}x)^2-(\mathrm{Sin}^{-1}x)^2\le\frac{3\pi^2}{4}$

W.E-6 : ***The set of values of x such that*** $\mathrm{Sin}^{-1}x-\mathrm{Cos}^{-1}x>0$ ***are***

Sol: $\mathrm{Sin}^{-1}x-\mathrm{Cos}^{-1}x>0 \Rightarrow \mathrm{Sin}^{-1}x>\frac{\pi}{4}$

$\therefore\ x\in\left(\frac{1}{\sqrt{2}},1\right]$ *as* $|x|\le 1$

W.E-7: The sum to the n terms of the series

$Co\sec^{-1}\sqrt{10}+Co\sec^{-1}\sqrt{50}+Co\sec^{-1}\sqrt{70}+.....$

$...+Co\sec^{-1}\sqrt{(n^2+1)(n^2+2n+2)}$ ***is***

Sol: Let $y=Co\sec^{-1}\sqrt{(n^2+1)(n^2+2n+2)}$

$Co\sec^2 y=(n^2+1)\left[(n^2+1)+2n+1\right]$

$=(n^2+1)^2+2n(n^2+1)+n^2+1$

$=(n^2+n+1)^2+1$

$\cot^2 y=(n^2+n+1)^2 \quad \left[\because \cos ec^2\theta-\cot^2\theta=1\right]$

$Tany=\frac{1}{n^2+n+1}=\frac{(n+1)-n}{1+(n+1)n}$

$y=Tan^{-1}\left[\frac{(n+1)-n}{1+(n+1)n}\right]$

$y=Tan^{-1}(n+1)-Tan^{-1}n$

Thus sum of n terms of the given series

$y=(Tan^{-1}2-Tan^{-1}1)+(Tan^{-1}3-Tan^{-1}2)+$
$(Tan^{-1}4-Tan^{-1}3)+......+(Tan^{-1}(n+1)-Tan^{-1}n)$

$=Tan^{-1}(n+1)-Tan^{-1}1 \quad =Tan^{-1}(n+1)-\frac{\pi}{4}$

W.E-8: $Sin^{-1}(\sin 5)=$

Sol: Here $\theta=5$ rad, Clearly it does not lie between $\frac{-\pi}{2}$ *and* $\frac{\pi}{2}$. But $2\pi-5$ and $5-2\pi$ both lies between $\frac{-\pi}{2}$ *and* $\frac{\pi}{2}$.

$Sin(5-2\pi)=Sin\left[-(2\pi-5)\right]=-Sin(2\pi-5)=Sin5$

$\therefore Sin^{-1}(Sin5)=Sin^{-1}(Sin(5-2\pi))=5-2\pi$

W.E-9 : $Cos^{-1}(\cos 10)$ ***is equal to....***

Sol: We know that $Cos^{-1}(\cos\theta)=\theta$ if $0\le\theta\le\pi$

Here $\theta=10\,rad$, clearly, it does not lie between 0 and π. But $4\pi-10$ lies between 0 and π

$\therefore Cos^{-1}(\cos 10)=Cos^{-1}(\cos(4\pi-10))=4\pi-10$

W.E-10: $Tan^{-1}(Tan(-6))$ ***is equal to.....***

Sol: We know that $Tan^{-1}(Tan\theta)=\theta$ if $\frac{-\pi}{2}<\theta<\frac{\pi}{2}$

Here $\theta=-6\,rad$, does not lie between $\frac{-\pi}{2}$ *and* $\frac{\pi}{2}$

But $2\pi-6$ lies between $\frac{-\pi}{2}$ *and* $\frac{\pi}{2}$

Now $Tan(2\pi-6)=-Tan6=Tan(-6)$

$\therefore Tan^{-1}(Tan(-6))=Tan^{-1}(Tan(2\pi-6))=2\pi-6$

C.U.Q

1. **$\tan(\cos^{-1}x) =$**

1) $\frac{\sqrt{1-x^2}}{x}$ 2) $\frac{x}{1-x^2}$ 3) $\frac{\sqrt{1+x^2}}{x}$ 4) $\sqrt{1-x^2}$

2. **The value of $\sin\left(\cot^{-1}\left(\cos\left(\tan^{-1}x\right)\right)\right)$ is**

1) $\sqrt{\frac{x^2+2}{x^2+1}}$ 2) $\sqrt{\frac{x^2+1}{x^2+2}}$

3) $\frac{x}{\sqrt{x^2+2}}$ 4) $\frac{1}{\sqrt{x^2+2}}$

3. $\tan\left[2\tan^{-1}(\cos x)\right] =$

1) 2tanx cosx 2) 2tanx cosecx

3) 2cotx cosecx 4) 2secx

4. $\cos^2\left[\tan^{-1}\left(\sin(\cot^{-1}x)\right)\right] =$

1) $\frac{x^2-1}{x^2+2}$ 2) $\frac{x^2+1}{x^2-2}$ 3) $\frac{x^2+1}{x^2+2}$ 4) $\frac{x^2-1}{x^2+1}$

5. **If a> b> c>0 , then**

$\cot^{-1}\left(\frac{ab+1}{a-b}\right)+\cot^{-1}\left(\frac{bc+1}{b-c}\right)+\cot^{-1}\left(\frac{ca+1}{c-a}\right)$

1) 0 2) $\pi/2$ 3) π 4) $3\pi/2$

6. **If** $\sec^{-1}\sqrt{1+x^2}+\operatorname{cosec}^{-1}\frac{\sqrt{1+y^2}}{y}+\cot^{-1}\frac{1}{z}=\pi$,

then

1) x + y + z = 0 2) x + y + z = 1

3) x + y + z = xyz 4) x + y + z = –xyz

7. **$\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right)-\frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] =$**

1) 0 2) 1 3) $\frac{x-y}{1+xy}$ 4) $\frac{2x}{1-x^2}$

8. **$\cos[\sin^{-1}(2\cos^2\theta - 1) + \cos^{-1}(1 - 2\sin^2\theta)] =$**

1) 0 2) 1 3) -1 4) $\frac{\pi}{2}$

9. **If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$ then**

(Eam-2010)

1) xy + yz + zx = 1 2) $x^2 + y^2 + z^2 + 2xyz = 1$

3) x + y + z = xyz 4) $\sum x+\sum yz=1+xyz$

10. $\cos^{-1}(x/a) + \cos^{-1}(y/b) = \theta$ **then**

$\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\theta+\frac{y^2}{b^2}=$

1) $\sin^2\theta$ 2) $\cos^2\theta$ 3) $\tan^2\theta$ 4) $\cot^2\theta$

11. $2\sin^{-1}x = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ **holds good for**

1) $x \in [0, 1]$ 2) $x \in [-1, 1]$

3) $x \in \left[-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right]$ 4) $x \in [-1, 0]$

12. $\operatorname{cosec}^{-1}(\cos x)$ **is defined if**

1) $x\in[-1, 1]$ 2) $x \in R$

3) $x-(2n\pi+1)\frac{\pi}{2},\ n\in z$ 4) $x=n\pi,\ n\in z$

13. **If $0<x<1$ then $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{1+x}\right)$ is equal to**

1) $\frac{1}{2}\cos^{-1}x$ 2) $\cos^{-1}\sqrt{\frac{1+x}{2}}$

3) $\sin^{-1}\sqrt{\frac{1-x}{2}}$ 4) All the above

C.U.Q-KEY

01) 1	02) 2	03) 3	04) 3	05) 3
06) 3	07) 3	08) 1	09) 1	10) 1
11) 3	12) 4	13) 4		

C.U.Q-HINTS

1. $\cos^{-1}x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

2. $\tan^{-1}x = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

3. Apply $2\tan^{-1}x$ formula

4. $\cot^{-1}x=\sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ & $\tan^{-1}x=\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

5. $\cot^{-1}(-x)=\pi-\cot^{-1}x$

6. Put x = tanA, y = tanB, z = tanC

7. Put x = tanA ; y = tanB

8. $2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \cos 2\theta$

9. Put $x = \frac{1}{\sqrt{3}} = y = z$

10. Apply $\cos^{-1}x + \cos^{-1}y$ formula

11. Standard $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

12. $Cosec^{-1}(\cos x)$ is defined only if $\cos x = \pm 1$
$\Rightarrow x = n\pi; \quad n \in z$

13. Put $x = \frac{1}{2}$, $G.E = \frac{\pi}{6}$

LEVEL - I (C.W)

1. The domain of $\text{Sin}^{-1}x + \text{Cos}^{-1}x$ is

1) $(-\pi, \pi)$ 2) $[-1,1]$ 3) $(0, 2\pi)$ 4) $(-\infty, \infty)$

2. The domain of $\log_e \sin^{-1}(x)$ is

1) (0, 1] 2) (0, 2] 3) $(0, \infty)$ 4) $(-\infty, 0]$

3. The range of $\text{Tan}^{-1}x$ is

1) R 2) $(0, \pi)$ 3) $[0, \pi]$ 4) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

4. The principal value of $\text{Cos}^{-1}\left(\cos\frac{7\pi}{6}\right)$ is

1) $\frac{7\pi}{6}$ 2) $\frac{5\pi}{3}$ 3) $\frac{5\pi}{6}$ 4) $\frac{13\pi}{6}$

5. $Sin^{-1}(\frac{\sqrt{3}}{2}) - Tan^{-1}(-\sqrt{3})$ **is**

1) $\frac{-\pi}{3}$ 2) $\frac{2\pi}{3}$ 3) $\frac{\pi}{6}$ 4) 0

6. $\cot^{-1}\left(2+\sqrt{3}\right) =$

1) $\frac{\pi}{12}$ 2) $\frac{\pi}{15}$ 3) $\frac{\pi}{5}$ 4) $\frac{3\pi}{10}$

7. $\cot\left[\text{Sin}^{-1}\sqrt{\frac{13}{17}}\right] - \sin\left[\text{Tan}^{-1}\frac{2}{3}\right] =$

1) $-\frac{2}{\sqrt{13}}$ 2) 0 3) $\frac{2}{\sqrt{13}}$ 4) $\frac{2}{3\sqrt{13}}$

8. $\sec^2(\cot^{-1}½) + \text{cosec}^2(\tan^{-1} 1/3) =$

1) 5 2) 10 3) 15 4) 50

9. Find the value of $\text{Sin}\left(\text{Sin}^{-1}\frac{4}{5} + \text{Sin}^{-1}\frac{7}{25}\right)$

1) $\frac{119}{125}$ 2) $\frac{117}{125}$ 3) $\frac{118}{125}$ 4) $\frac{113}{125}$

10. $\cos(2\cos^{-1}(7/25)) =$

1) $\frac{527}{625}$ 2) $-\frac{527}{625}$ 3) $\pi - \frac{527}{625}$ 4) $\frac{24}{25}$

11. If $x + \frac{1}{x} = 2$, the principle value of $\text{Sin}^{-1}x$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) $\frac{3\pi}{2}$

12. The numerical value of $\tan\left(2\,\text{Tan}^{-1}\frac{1}{5} - \frac{\pi}{4}\right)$ **is**

1) 1 2) 0 3) $\frac{7}{17}$ 4) $-\frac{7}{17}$

13. $2\,\text{Tan}^{-1}(1/2) + \text{Sin}^{-1}(3/5) =$

1) $\tan^{-1}(\frac{12}{25})$ 2) $\frac{\pi}{4}$

3) $\frac{\pi}{2}$ 4) $\tan^{-1}(\frac{25}{12})$

14. $\text{Tan}^{-1}(2) + \text{Tan}^{-1}(3) =$

1) $-\frac{\pi}{4}$ 2) $\frac{\pi}{4}$ 3) $\frac{3\pi}{4}$ 4) $\frac{5\pi}{4}$

15. $Tan^{-1}(\frac{m}{n}) - Tan^{-1}(\frac{m-n}{m+n}) =$

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

16. $2\,\text{Tan}^{-1}\frac{1}{3} + \text{Tan}^{-1}\frac{1}{7} =$

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

17. $\sec\left[\text{Tan}^{-1}5 + \text{Tan}^{-1}\frac{1}{5} - \text{Tan}^{-1}\frac{3}{4}\right] =$

1) 3/5 2) 5/3 3) 4/5 4) $\sqrt{2}$

18. $4\,\text{Tan}^{-1}\frac{1}{5} - \text{Tan}^{-1}\frac{1}{70} + \text{Tan}^{-1}\frac{1}{99} =$

1) π 2) $\pi/2$ 3) $\pi/4$ 4) $3\pi/4$

19. $Tan^{-1}\left(\frac{5}{12}\right) + Sin^{-1}\left(\frac{24}{25}\right) = Cos^{-1}(x) \Rightarrow x =$

1) $\frac{-31}{325}$ 2) $\frac{-33}{325}$ 3) $\frac{-36}{325}$ 4) $\frac{-39}{325}$

20. **If** $\text{Sin}^{-1}(\frac{12}{13}) + \text{Sec}^{-1}(\frac{13}{x}) = \frac{\pi}{2}$ **then x =**

1) 12 2) 13 3) 11 4) 5

21. **If** $\text{Sin}^{-1}(\frac{3}{x}) + \text{Sin}^{-1}(\frac{4}{x}) = \frac{\pi}{2}$ **then x = (EAM-2008)**

1) 6 2) 7 3) 8 4) 5

22. $Tan^{-1}(\frac{a}{x}) + Tan^{-1}(\frac{b}{x}) = \frac{\pi}{2}$ *then* $x =$

1) ab 2) $\sqrt{ab}$ 3) $\sqrt{a^2+b^2}$ 4) a^2+b^2

23. $Tan^{-1}\left(x+\sqrt{1+x^2}\right) =$

1) $\frac{\pi}{4} - \frac{1}{2}\tan^{-1}x$ 2) $\frac{1}{2}\tan^{-1}x$

3) $\frac{\pi}{2} - \frac{1}{2}\tan^{-1}x$ 4) $\frac{\pi}{4} + \frac{1}{2}\tan^{-1}x$

24. **If** $\text{Tan}^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right) = k\,\text{Tan}^{-1}\left(\frac{x}{a}\right)$ **then k =**

1) 2 2) 3 3) - 2 4) 4

25. **If** $\text{Tan}^{-1}\left(\frac{2x}{x^2-1}\right) + \text{Cos}^{-1}\frac{x^2-1}{x^2+1} = \frac{2\pi}{3}$ **then x =**

1) $2 - \sqrt{3}$ 2) $\sqrt{3} - \sqrt{2}$ 3) $2 + \sqrt{3}$ 4) $+\sqrt{2}$

26. **If** $\text{Tan}^{-1}\left(\frac{x-1}{x-2}\right) + \text{Tan}^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ **then x=**

1) $\frac{1}{\sqrt{2}}$ 2) $\pm\frac{1}{\sqrt{2}}$ 3) $\pm\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

27. **If** $\text{Cos}^{-1}x = \text{Cot}^{-1}(4/3) + \text{Tan}^{-1}(1/7)$ **then x =**

1) ½ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{3}{5}$

28. **The trigonometric equation** $\text{Sin}^{-1}x = 2\text{Sin}^{-1}a$ **has a solution for (AIEEE-2003)**

1) $|a| \le \frac{1}{\sqrt{2}}$ 2) all real values of a

3) $|a| < \frac{1}{2}$ 4) $|a| \ge \frac{1}{\sqrt{2}}$

29. **If n** $\in$ **N,** $\sum_{k=1}^{n} Sin^{-1}(x_k) = \frac{n\pi}{2}$ **then** $\sum_{k=1}^{n} x_k =$

1) n 2) k 3) $\frac{k(k+1)}{2}$ 4) $\frac{n(n+1)}{2}$

30. **If** $\text{Sin}^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots\right) + \text{Cos}^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\right) = \frac{\pi}{2}$ **for** $0 < |x| < \sqrt{2}$ **then x =**

1) 1/2 2) 1 3) - 1/2 4) -1

31. **The value of** $\text{Sin}^{-1}(\sin 10)$ **is**

1) 10 2) $10-3\pi$ 3) $3\pi-10$ 4) -10

32. **The greatest of** $\text{Tan}^{-1}1, \text{Sin}^{-1}1, \sin 1, \cos 1$ **is**

1) $\sin 1$ 2) $\cos 1$ 3) $\tan^{-1}1$ 4) $\sin^{-1}1$

33. **If** $\text{Tan}^{-1}x, \text{Tan}^{-1}y, \text{Tan}^{-1}z$ **are in A.P. then** $\frac{2y}{1-y^2} =$

1) $\frac{x-z}{1+xz}$ 2) $\frac{x+z}{1-xz}$ 3) $x+z$ 4) xz

LEVEL - I (C.W)-KEY

01) 2	02) 1	03) 4	04) 3	05) 2
06) 1	07) 2	08) 3	09) 2	10) 2
11) 2	12) 4	13) 3	14) 3	15) 3
16) 3	17) 2	18) 3	19) 3	20) 1
21) 4	22) 2	23) 4	24) 2	25) 1
26) 2	27) 3	28) 1	29) 1	30) 2
31) 3	32) 4	33) 2		

LEVEL - I (C.W)-HINTS

1. Domain $=[-1, 1]$
2. Domain of $\log_e x$ is $x > 0$
3. Range of $\text{Tan}^{-1}x = \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

4. $Cos^{-1}[Cos(\frac{7\pi}{6})]=Cos^{-1}[\cos(2\pi-\frac{5\pi}{6})]=\frac{5\pi}{6}$

5. $Sin^{-1}(\frac{\sqrt{3}}{2})-Tan^{-1}(-\sqrt{3})=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$

6. $Cot15^0=2+\sqrt{3}$

7. $Sin^{-1}\sqrt{\frac{13}{17}}=Cot^{-1}\frac{2}{\sqrt{13}}$, $Tan^{-1}\frac{2}{3}=Sin^{-1}\frac{2}{\sqrt{13}}$

8. $\sec^2\left(\tan^{-1}2\right)+\operatorname{cosec}^2\left(\cot^{-1}3\right)=$

$1+\tan^2\left(\tan^{-1}2\right)+1+\cot^2\left(\cot^{-1}3\right)$

9. Apply $Sin^{-1}x+Sin^{-1}y=Sin^{-1}[x\sqrt{1-y^2}+y\sqrt{1-x^2}]$ formula where $x=\frac{4}{5}$, $y=\frac{7}{25}$

10. Apply $2Cos^{-1}x=Cos^{-1}(2x^2-1)$ formula where $x=\frac{7}{25}$

11. Put X=1

12. Apply $2Tan^{-1}x=Tan^{-1}(\frac{2x}{1-x^2})$ formula where X= 1/5

13. $2Tan^{-1}\left(\frac{1}{2}\right)=Cos^{-1}\left(\frac{1-\frac{1}{4}}{1+\frac{1}{4}}\right)=Cos^{-1}\left(\frac{3}{5}\right)$

14. $Tan^{-1}x+Tan^{-1}y=\pi+Tan^{-1}(\frac{x+y}{1-xy})$, x>0,y>0 xy>1 formula

15. Apply $Tan^{-1}x-Tan^{-1}y=Tan^{-1}(\frac{x-y}{1+xy})$, formula

16. Apply $2Tan^{-1}x$ formula after

$\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)$

17. $Tan^{-1}5+Tan^{-1}\frac{1}{5}=\frac{\pi}{2}$ and $Tan^{-1}\frac{3}{4}=Co\sec^{-1}\frac{5}{3}$

18. $4Tan^{-1}\frac{1}{5}=Tan^{-1}\frac{120}{119}$ Apply $Tan^{-1}x-Tan^{-1}y$ and $Tan^{-1}x+Tan^{-1}y$

19. $Sin^{-1}(\frac{24}{25})=Tan^{-1}(\frac{24}{7})$ and apply

$Tan^{-1}x+Tan^{-1}y=\pi+Tan^{-1}\left(\frac{x+y}{1-xy}\right)$; $xy>1$

20. $Sec^{-1}x=Cos^{-1}\frac{1}{x}$ and $Sin^{-1}x+Cos^{-1}x=\frac{\pi}{2}$

21. Apply $Sin^{-1}\frac{a}{x}+Sin^{-1}\frac{b}{x}=\frac{\pi}{2}$ then $x=\sqrt{a^2+b^2}$

22. Apply $Tan^{-1}x+Cot^{-1}x=\frac{\pi}{2}\Rightarrow\frac{a}{x}=\frac{x}{b}\Rightarrow x=\sqrt{ab}$

23. Put $x=Tan\theta$

24. Put $\frac{x}{a}=Tan\theta$

25. Put $x=Tan\theta$

26. Apply $Tan^{-1}x+Tan^{-1}y$ formula

27. $Cos^{-1}x=Tan^{-1}\frac{3}{4}+Tan^{-1}(\frac{1}{7})$ Apply

$Tan^{-1}\left(\frac{p}{q}\right)+Tan^{-1}\left(\frac{q-p}{q+p}\right)=\frac{\pi}{4}$

$\Rightarrow Cos^{-1}x=\frac{\pi}{4}\Rightarrow x=\frac{1}{\sqrt{2}}$

28. The range of $Sin^{-1}x=[\frac{-\pi}{2},\frac{\pi}{2}]$

$\therefore -\frac{\pi}{2}\le Sin^{-1}x\le\frac{\pi}{2}$

$\Rightarrow -\frac{\pi}{2}\le 2Sin^{-1}a\le\frac{\pi}{2}$ $\Rightarrow -\frac{\pi}{4}\le Sin^{-1}a\le\frac{\pi}{4}$

$\Rightarrow -\frac{1}{\sqrt{2}}\le a\le\frac{1}{\sqrt{2}}\Rightarrow|a|\le\frac{1}{\sqrt{2}}$

29. Put $x_1=x_2=x_3=........=1$

$$\sum_{k=1}^{n}x_k=1+1+1+........+1(n\ time)=n$$

30. Put $x=x^2\Rightarrow x(x-1)=0\Rightarrow x=0\ (or)\ 1$

31. radian$=\frac{7\pi}{22}=57^0 17^1 44.8^{11}$ and Sin^{-1} lies

between $-\frac{\pi}{2}$ *and* $\frac{\pi}{2}$

32. 1 radian= $=57^{0}17^{1}44.8^{11}$ $Sin^{-1}1=\frac{\pi}{2}$ is

the greatest

33. $2Tan^{-1}y=Tan^{-1}x+Tan^{-1}z$

$Tan^{-1}(\frac{2y}{1-y^{2}})=Tan^{-1}(\frac{x+z}{1-xz})$

LEVEL - I (H.W)

1. The domain of $Sin^{-1}\frac{2x+1}{3}$ **is**

1. (-2, 1] 2. [-2, 1] 3. R 4. [-1, 1]

2. The domain of $Cos^{-1}\sqrt{2x}$ **is**

1) [-1, 1] 2) [-1/2, 1/2]
3) [0, 1/2] 4) (1, 1/2)

3. The range of $Cot^{-1}x$ **isF**

1) $(0,\pi)$ 2) $[0,\pi]$ 3) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ 4) R

4. The principal value of $Sin^{-1}\left(Sin\left(\frac{2\pi}{3}\right)\right)$ **is**

1) $\frac{2\pi}{3}$ 2) $\frac{\pi}{3}$ 3) $\frac{-\pi}{3}$ 4) $\frac{-2\pi}{3}$

5. The principal value of $Sin^{-1}(Tan(\frac{-5\pi}{4}))$ **is**

1) $\frac{\pi}{4}$ 2) $\frac{-\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{-\pi}{2}$

6. $Sec^{-1}(\frac{2\sqrt{2}}{1+\sqrt{3}})$

1) $\frac{\pi}{12}$ 2) $\frac{\pi}{3}$ 3) $\frac{3\pi}{4}$ 4) $\frac{\pi}{6}$

7. The value of $Sin(2Tan^{-1}\frac{1}{3})+Cos(Tan^{-1}2\sqrt{2})$ **is**

1) $\frac{12}{13}$ 2) $\frac{13}{14}$ 3) $\frac{14}{15}$ 4) $\frac{16}{15}$

8. $Sec^2(Tan^{-1}(2))+Cosec^2(Cot^{-1}(2))=$

1) 5 2) 10 3) 15 4) 50

9. $Cos^{-1}(\frac{4}{5})+Cos^{-1}(\frac{63}{65})=$

1) $Cos^{-1}(\frac{204}{325})$ 2) $Cos^{-1}(\frac{300}{325})$

3) $Cos^{-1}(\frac{201}{300})$ 4) $Sin^{-1}(\frac{204}{325})$

10. sin (1/2 Cot^{-1} (-3/4)) =

1) $1/\sqrt{5}$ 2) $2/\sqrt{5}$ 3) $-2/\sqrt{5}$ 4) $-1/\sqrt{5}$

11. If '*x*' is a negative real number then the value of $Cos^{-1}x+Sec^{-1}x$ is

1) rational number 2) irrational number
3) integer 4) imaginary

12. $\tan(2\,Tan^{-1}(\frac{\sqrt{5}-1}{2}))=$

1) 4 2) 3 3) $\frac{1}{2}$ 4) 2

13. $Sin^{-1}\frac{4}{5}+2Tan^{-1}\frac{1}{3}=$ **(Eam-2005)**

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

14. $Tan^{-1}(\frac{1}{2})+Tan^{-1}(\frac{1}{3})=$ **(Eam-1999)**

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

15. $Tan^{-1}(\frac{3}{2})-Tan^{-1}(\frac{1}{5})=$

1) $\frac{-\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

16. $4\,Tan^{-1}\frac{1}{5}-Tan^{-1}\frac{1}{239}=$

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

17. $Cot(\operatorname{cosec}^{-1}\frac{5}{3}+Tan^{-1}\frac{2}{3})=$

1) $\frac{6}{17}$ 2) $\frac{3}{17}$ 3) $\frac{4}{17}$ 4) $\frac{5}{17}$

18. $2Tan^{-1}\frac{1}{5}+Tan^{-1}\frac{1}{7}+2Tan^{-1}\frac{1}{8}=$

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

19. **If** $Sin^{-1}x - Cos^{-1}x = \frac{\pi}{6}$ **then x = (Eam-2002)**

1) 1/2 2) $\frac{\sqrt{3}}{2}$ 3) - 1/2 4) $-\frac{\sqrt{3}}{2}$

20. $Cos^{-1}(\sqrt{3}x)+Cos^{-1}x=\frac{\pi}{2}$ **then x =**

1) 0 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{1}{2}$ 4) 1

21. $Sin^{-1}(\frac{5}{x})+Sin^{-1}(\frac{12}{x})=\frac{\pi}{2}$ **, (x > 0) then x =**

1) 12 2) 13 3) 14 4) 15

22. $Sin^{-1}(\frac{x}{5})+Cosec^{-1}(\frac{5}{4})=\frac{\pi}{2}$ **then x = (AIE-2007)**

1) 1 2) 3 3) 4 4) 5

23. $Tan^{-1}(\frac{1+\sin x}{\cos x})=$

1) $\frac{\pi}{4}-\frac{x}{2}$ 2) $\frac{\pi}{4}-x$ 3) $\frac{\pi}{4}+x$ 4) $\frac{\pi}{4}+\frac{x}{2}$

24. **A value of** $Tan^{-1}\left\{\sin\left(Cos^{-1}\sqrt{\frac{2}{3}}\right)\right\}$ **is**

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

25. $Tan[\frac{1}{2}Sin^{-1}(\frac{2a}{1+a^2})+\frac{1}{2}Cos^{-1}(\frac{1-a^2}{1+a^2})]=$

1) $\frac{2a}{1+a^2}$ 2) $\frac{2a}{1-a^2}$ 3) $\frac{a}{1+a^2}$ 4) $\frac{a}{1-a^2}$

26. **If** $3Tan^{-1}(2-\sqrt{3})-Tan^{-1}(x)=Tan^{-1}(1/3)$ **then x =**

1) 1/2 2) 2 3) 3 4) 1/3

27. **If** $Cos^{-1}(\frac{8}{17})-Cos^{-1}(\frac{5}{13})=Cos^{-1}x$ **then x =**

1) $\frac{140}{221}$ 2) $\frac{220}{221}$ 3) $\frac{7}{11}$ 4) $\frac{221}{220}$

28. **The equation** $2Cos^{-1}x=Cos^{-1}(2x^2-1)$ **is satisfied by**

1) $-1\le x\le 1$ 2) $0\le x\le 1$
3) $x\ge 1$ 4) $x\le 1$

29. **If** $\sum_{r=1}^{n}Cos^{-1}x_r=0$**, then** $\sum_{r=1}^{n}x_r$ **equals to**

1) 0 2) n 3) $\frac{n(n+1)}{2}$ 4) $\frac{n}{2}$

30. $Sin^{-1}\left[x\sqrt{1-x}-\sqrt{x}\sqrt{1-x^2}\right]=$

1) $\sin^{-1}x+\sin^{-1}\sqrt{x}$ 2) $\sin^{-1}x-\sin^{-1}\sqrt{x}$
3) $\sin^{-1}\sqrt{x}-\sin^{-1}x$ 4) 0

31. **If** $\theta\in\left[\frac{\pi}{2},\frac{3\pi}{2}\right]$ **then** $Sin^{-1}(\sin\theta)=$

1) θ 2) $\pi-\theta$ 3) $2\pi-\theta$ 4) $-\pi+\theta$

32 **$Cos^{-1}(\frac{a-b}{a+b})=$**

1) $2\,Tan^{-1}b$ 2) $2\,Tan^{-1}\sqrt{\frac{b}{a}}$

3) $2\,Tan^{-1}\sqrt{\frac{a}{b}}$ 4) $2\,Tan^{-1}(\frac{ab}{a-b})$

33. **If** $\theta=Tan^{-1}a,\ \phi=Tan^{-1}b$ **and ab=-1 , then** $\theta-\phi$ **is equal to :**

1) 0 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) π

LEVEL - I (H.W)-KEY

01) 2	02) 3	03) 1	04) 2	05) 4
06) 1	07) 3	08) 2	09) 1	10) 2
11) 2	12) 4	13) 2	14) 2	15) 3
16) 3	17) 1	18) 3	19) 2	20) 3
21) 2	22) 2	23) 4	24) 4	25) 2
26) 1	27) 2	28) 2	29) 2	30) 2
31) 2	32) 2	33) 3		

LEVEL - I (H.W)-HINTS

1. $Sin^{-1}x$ domain [-1, 1] $\Rightarrow -1\le\frac{2x+1}{3}\le 1$

2. $0 \le 2x \le 1$

3. Range of $\cot^{-1} x$ is $(0, \pi)$

4. $Sin^{-1}(Sin\left(\frac{2\pi}{3}\right) = Sin^{-1}[Sin(\pi - \frac{\pi}{3})]$

5. $Sin^{-1}[Tan\left(\frac{-5\pi}{4}\right)] = Sin^{-1}(-1)$

6. $\frac{2\sqrt{2}}{1+\sqrt{3}} = Sec\frac{\pi}{12}$

7. Apply $2Tan^{-1}x = Sin^{-1}(\frac{2x}{1+x^2})$ and

$Tan^{-1}x = Cos^{-1}(\frac{1}{\sqrt{1+x^2}})$ formula

8. $Sec^2(Tan^{-1}2) + Co\sec^2\left(Cot^{-1}2\right)$

$=1+Tan^2(Tan^{-1}2)+1+Cot^2(Cot^{-1}2)$

9. Apply

$Cos^{-1}x + Cos^{-1}y = Cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$

formula

10. Put $Cot^{-1}(\frac{-3}{4}) = x$

11. $x = -1$ only satisfied

$\therefore Cos^{-1}x + Sec^{-1}x = 2\pi$ =Irrational number

12. Apply $2Tan^{-1}x$ formula

13. Apply $2Tan^{-1}x$ formula

$\therefore Sin^{-1}\frac{4}{5} + 2Tan^{-1}\frac{1}{3} = Sin^{-1}\frac{4}{5} + Cos^{-1}\frac{4}{5} = \frac{\pi}{2}$

14. Apply $Tan^{-1}(\frac{p}{q}) + Tan^{-1}\left(\frac{q-p}{q+p}\right) = \frac{\pi}{4}$

15. Apply $Tan^{-1}x - Tan^{-1}y$

16. $4Tan^{-1}\left(\frac{1}{5}\right) = Tan^{-1}(\frac{120}{19})$ then apply

$Tan^{-1}x - Tan^{-1}y$ formula

17. $Cot^{-1}(Co\sec^{-1}\frac{5}{3} + Tan^{-1}\frac{2}{3})$

$= Cot(Sin^{-1}(\frac{3}{5}) + Tan^{-1}(\frac{2}{3}))$

Apply $Sin^{-1}x = Tan^{-1}\frac{x}{\sqrt{1-x^2}}$ and

$\Rightarrow Sin^{-1}\frac{3}{5} = Tan^{-1}\frac{3}{4}$ apply $Tan^{-1}x + Tan^{-1}y$

formula

18. Apply $2Tan^{-1}x = Tan^{-1}(\frac{2x}{1-x^2})$ and

$Tan^{-1}x + Tan^{-1}y$ formula

19. Put $x = \frac{\sqrt{3}}{2}$

20. Put $x = \frac{1}{2}$

21. Apply $Sin^{-1}(\frac{a}{x}) + Sin^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$ then

$x = \sqrt{a^2+b^2}$ formula

22. $Sin^{-1}\frac{x}{5} + Sin^{-1}\left(\frac{4}{5}\right) = \frac{\pi}{2}$

Apply $Sin^{-1}\frac{a}{x} + Sin^{-1}\frac{b}{x} = \frac{\pi}{2}$ then

$x = \sqrt{a^2+b^2} \Rightarrow 5 = \sqrt{x^2+4^2} \Rightarrow x = 3$

23. $\frac{1+Sinx}{Cosx} = Tan(\frac{\pi}{4} + \frac{x}{2})$

24. $Tan^{-1}[Sin(Sin^{-1}\sqrt{\frac{1}{3}})] = Tan^{-1}(\frac{1}{\sqrt{3}}) = \frac{\pi}{6}$

25. Put $a = Tan\theta$

26. $2-\sqrt{3} = Tan15^0$ then $Tan^{-1}x = \frac{\pi}{4} - Tan^{-1}\frac{1}{3}$

27. Apply $Cos^{-1}x - Cos^{-1}y = Cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$

28. Given equation satisfied for $0 \le x \le 1$

29. $Cos^{-1}x_1 = Cos^{-1}x_2 = ... = Cos^{-1}x_n = 0$

$\Rightarrow x_1 = x_2 = ... = x_n = +1$

$$\therefore \sum_{r-1}^{n} x_r = n$$

30. $Sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}] = Sin^{-1}[x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2}]$

$= Sin^{-1}x - Sin^{-1}\sqrt{x}$

31. $Sin^{-1}(Sin\theta) = Sin^{-1}(Sin(\pi-\theta)) = \pi - \theta$

32. $Cos^{-1}\left(\dfrac{1-\frac{b}{a}}{1+\frac{b}{a}}\right) = Cos^{-1}\left(\dfrac{1-\left(\sqrt{\frac{b}{a}}\right)^2}{1+(\sqrt{\frac{b}{a}})^2}\right) = 2Tan^{-1}\sqrt{\dfrac{b}{a}}$

33. $Tan(\theta-\Phi) = \dfrac{Tan\theta - Tan\Phi}{1+Tan\theta Tan\Phi} = \dfrac{a-b}{1-1} = \infty$

$\theta - \Phi = \dfrac{\pi}{2}$

LEVEL - II (C.W)

1. Range of $\mathrm{Sin^{-1}x + Cos^{-1}x + Tan^{-1}x}$ **is**

1) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ 2) $(0, \pi]$ 3) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ 4) $[0, \pi]$

2. Range of $f(x) = \mathrm{Sin}^{-1}x + \mathrm{Tan}^{-1}x + \mathrm{Sec}^{-1}x$ **is**

1) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ 2) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

3) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ 4) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

3. The domain of $\mathrm{Sin^{-1}[\log_2(x^2/2)]}$ **is**

1) $[-2,-1]$ 2) $[1, 2]$

3) $[-2,-1] \cup [1, 2]$ 4) $[-2,0]$

4. Arrange the following functions in the increasing order of their domains.

$f(x) = \mathrm{Sin}^{-1}x$; $g(x) = \mathrm{Cos}^{-1}\sqrt{2x}$; $h(x) = \mathrm{Sin}^{-1}2x$; $k(x) = \mathrm{Tan}^{-1}(5x)$

1) $f(x), g(x), h(x), k(x)$ 2) $g(x), h(x), k(x), f(x)$

3) $g(x), h(x), f(x), k(x)$ 4) $g(x), f(x), h(x), k(x)$

5. The value of $\cos(2\mathrm{Cos}^{-1} 0.8)$ **is**

1) 0.48 2) 0.96 3) 0.6 4) 0.28

6. The value of the expression

$Sin^{-1}\left(Sin\frac{22\pi}{7}\right) + Cos^{-1}\left(Cos\frac{5\pi}{3}\right) + Tan^{-1}\left(Tan\frac{5\pi}{7}\right) + Sin^{-1}(Cos2)$

is

1) $\frac{17\pi}{42} - 2$ 2) -2 3) $-\frac{\pi}{21} - 2$ 4) $\frac{\pi}{21}$

7. $\cos\left[2\,\mathrm{Sin}^{-1}\sqrt{\frac{1-x}{2}}\right] =$

1) x 2) $\frac{1}{x}$ 3) 2x 4) 3x

8. If $\mathrm{Sin^{-1}(x) - Cos^{-1}(x) = Sin^{-1}(x-1)}$ **then x =** **[EAM-2004]**

1) 0, ½ 2) -1, ½ 3) 1, - ½ 4) ½, 1

9. If $\mathrm{Sin^{-1}(6x) + Sin^{-1}(6\sqrt{3}\,x)} = -\frac{\pi}{2}$ **then the value of x is**

1) $\frac{1}{12}$ 2) $-\frac{1}{12}$ 3) $-\frac{1}{4\sqrt{3}}$ 4) $\frac{1}{4\sqrt{3}}$

10. If $\mathrm{Tan^{-1}(2x) + Tan^{-1}(3x)} = \frac{\pi}{4}$ **then x =**

1) 1/2 2) 1/4 3) 1/6 4) 6

11. If $\mathrm{Cot^{-1}}\left(\frac{1-x^2}{2x}\right) + \mathrm{Cos^{-1}}\left(\frac{1-x^2}{1+x^2}\right) = \frac{2\pi}{3}$; **$x > 0$; $x \neq 1$ then x =**

1) $\frac{1}{\sqrt{2}}$ 2) $\pm\frac{1}{\sqrt{2}}$ 3) $\pm\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

12. If $Tan^{-1}\frac{1}{1+2x} + Tan^{-1}\frac{1}{4x+1} = Tan^{-1}\frac{2}{x^2}$ **then x =**

1) 1 2) 0 3) -3 4) $\frac{2}{3}$

13. $\mathrm{Tan}\left[\mathrm{Tan}^{-1}\left(\frac{1}{a+b}\right) + \mathrm{Tan}^{-1}\left(\frac{b}{a^2+ab+1}\right)\right] =$

1) a 2) 1/a 3) b 4) 1/b

14. In a ΔABC, **If C is a right angle then**

$Tan^{-1}\left(\dfrac{a}{b+c}\right) + Tan^{-1}\left(\dfrac{b}{c+a}\right) =$

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $\frac{5\pi}{2}$

15. **If** $\mathrm{Tan}^{-1}\left(\frac{x+1}{x-1}\right)+\mathrm{Tan}^{-1}\left(\frac{x-1}{x}\right)$ $= \pi + \mathrm{Tan}^{-1}(-7)$ **then** x =

1) 2 2) -2 3) 1 4) no solution

16. **The number of real solutions of**

$\mathrm{Tan}^{-1}\left(\sqrt{x(x+1)}\right)+\mathrm{Sin}^{-1}\sqrt{(x^2+x+1)}=\frac{\pi}{2}$ **is**

1) 0 2) 1 3) 2 4) ∞

17. **If x =** $\mathrm{Tan}^{-1}(1)+\mathrm{Cos}^{-1}(-\tfrac{1}{2})+\mathrm{Sin}^{-1}(-\tfrac{1}{2})$ **and** $y=\cos\left[\tfrac{1}{2}\mathrm{Cos}^{-1}(1/8)\right]$ **then**

1) $x=2\pi y$ 2) $y=3\pi x$

3) $x=\pi y$ 4) $y=\pi x$

18. **Let a,b,c be a positive real numbers**

$\theta=\mathrm{Tan}^{-1}\sqrt{\frac{a(a+b+c)}{bc}}+\mathrm{Tan}^{-1}\sqrt{\frac{b(a+b+c)}{ca}}+$ $\mathrm{Tan}^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ **then** $\tan\theta=$

1) 0 2) 3π 3) 1 4) 4π

19. **If** x, y **and** z **are in A.P and** $\mathrm{Tan}^{-1}x, \mathrm{Tan}^{-1}y$ **and** $\mathrm{Tan}^{-1}z$ **are also in A.P then (Aie-2013)**

1) $x=y=z$ 2) $2x=3y=6z$

3) $6x=3y=2z$ 4) $6x=4y=3z$

20. **If** $\left[\sin^{-1}\left(\cos^{-1}\left(\sin^{-1}\left(\tan^{-1}x\right)\right)\right)\right]=1$**, where** **[•] denotes the greatest integer function, then** $x\in$

1) $[\tan\sin\cos 1, \tan\sin\cos\sin 1]$

2) $(\tan\sin\cos 1, \tan\sin\cos\sin 1)$

3) $[-1, 1]$ 4) $[\sin\cos\tan 1, \sin\cos\sin\tan 1]$

21. **If** $x(3-x)\geq 2$ **then** $\mathrm{Sin}^{-1}(x)+\mathrm{Sin}^{-1}(x^2)+\ldots\ldots+\mathrm{Sin}^{-1}(x^{10})=$

1) $\frac{\pi}{2}$ 2) 2π 3) 5π 4) 10π

22. **If** $\mathrm{Cos}^{-1}x+\mathrm{Cos}^{-1}y=\frac{\pi}{2}$ **and** $\mathrm{Tan}^{-1}x-\mathrm{Tan}^{-1}y=0$ **then** $x^2+xy+y^2=$

1) 0 2) $-\frac{1}{2}$ 3) $\frac{1}{2}$ 4) $\frac{3}{2}$

23. **If** $0<x<1$**, then** $\sqrt{1+x^2}\left[\left\{x\cos(\cot^{-1}x)+\sin(\cot^{-1}x)\right\}^2-1\right]=$

1) $\frac{x}{\sqrt{1+x^2}}$ 2) x 3) $x\sqrt{1+x^2}$ 4) $\sqrt{1+x^2}$

24. **The value of** $'a'$ **for which** $ax^2+\mathrm{Sin}^{-1}(x^2-2x+2)+\mathrm{Cos}^{-1}(x^2-2x+2)=0$ **has a real solution, is**

1) $\frac{\pi}{2}$ 2) $-\frac{\pi}{2}$ 3) $\frac{2}{\pi}$ 4) $-\frac{2}{\pi}$

25. **If minimum value of** $\left(\mathrm{Sin}^{-1}x\right)^2+\left(\mathrm{Cos}^{-1}x\right)^2$ **is** $\frac{\pi^2}{k}$**, then value of k is**

1) 4 2) 6 3) 8 4) 10

26. **If** $\left(\mathrm{Tan}^{-1}x\right)^2+\left(Cot^{-1}x\right)^2=\frac{5\pi^2}{8}$**, then** $x=$ **[Eam-2011]**

1) -1 2) 1 3) 0 4) 2

27. **If** $6\,\mathrm{Sin}^{-1}(x^2-6x+12)=2\pi$**, then the value of** x **, is**

1) 1 2) 2 3) 3 4) does not exists

28. **If** $\mathrm{Cos}^{-1}x>\mathrm{Sin}^{-1}x$ **, then** x **belongs to the interval**

1) $(-\infty, 0)$ 2) $(-1, 0)$ 3) $[0, \frac{1}{\sqrt{2}})$ 4) $[-1, \frac{1}{\sqrt{2}})$

29. **The least integral value of** x **for which** $Tan^{-1}x>Cot^{-1}x$ **is**

1) 1 2) 2 3) 3 4) 4

30. **The value of** $\mathrm{Sin}^{-1}(\sin 12)+\mathrm{Sin}^{-1}(\cos 12)=$

1) 0 2) $24-2\pi$ 3) $4\pi-24$ 4) 8π

31. **If** $\mathrm{Cos}^{-1}x-\mathrm{Cos}^{-1}\left(\frac{y}{2}\right)=\alpha$ **then** $4x^2-4xy\cos\alpha+y^2=$ **[AIEEE-2005]**

1) $-4\sin^2\alpha$ 2) $4\sin^2\alpha$
3) 4 4) $2\sin 2\alpha$

32. **If** $\log_2^{x} \geq 0$ **then**

$\log_{\frac{1}{\pi}}\{Sin^{-1}\frac{2x}{1+x^2}+2Tan^{-1}x\}=$

1) $\log_{\frac{1}{\pi}}(4Tan^{-1}x)$ 2) 0 3) -1 4) -2

33. **If** $Sin^{-1}x+Sin^{-1}y+Sin^{-1}z=\frac{3\pi}{2}$ **then**

$\sum(\frac{x^{201}+y^{201}}{x^{603}+y^{603}})(\frac{x^{402}+y^{402}}{x^{804}+y^{804}})=$

1) 0 2) 1 3) 2 4) 3

34. **The ascending order of minimum values of the function** $P:Sin^{-1}x-Cos^{-1}x$

$Q:Tan^{-1}x+Cot^{-1}x$, $R:\sec^{-1}x-Co\sec^{-1}x$

1) P, Q, R 2) P, R, Q
3) Q, P, R 4) Q, R, P

35. $Tan^{-1}\left(\frac{1}{1+(1)(2)}\right)+Tan^{-1}\left(\frac{1}{1+(2)(3)}\right)+$

$\ldots\ldots\ldots\ldots\ldots+Tan^{-1}\left(\frac{1}{1+(n-1)(n)}\right)=$

1) 0 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $Tan^{-1}\left(\frac{n-1}{n+1}\right)$

LEVEL - II (C.W)-KEY

01) 3	02) 3	03) 3	04) 3	05) 4
06) 1	07) 1	08) 1	09) 2	10) 3
11) 4	12) 2	13) 2	14) 2	15) 1
16) 3	17) 3	18) 1	19) 1	20) 1
21) 3	22) 4	23) 3	24) 2	25) 3
26) 1	27) 4	28) 4	29) 2	30) 1
31) 2	32) 3	33) 4	34) 2	35) 4

LEVEL - II (C.W)-HINTS

1. Domain of $Sin^{-1}x+Cos^{-1}x+Tan^{-1}x$ is [-1,1]

Range is $[\frac{\pi}{2}+Tan^{-1}(-1),\ \frac{\pi}{2}+Tan^{-1}(1)]$

2. Common domain $=\{-1,1\}$

range $=\{f(-1),f(1)\}$

3. $-1\leq\log_2^{(\frac{x^2}{2})}\leq 1\Rightarrow 2^{-1}\leq\frac{x^2}{2}\leq 2\Rightarrow 1\leq x^2\leq 4$

4. f: $\sin^{-1}(x)$ domain $[-1,1]$

g: $\cos^{-1}\sqrt{2x}$ domain $\left[0,\frac{1}{2}\right]$

h: $\sin^{-1}2x$ domain $\left[\frac{-1}{2},\frac{1}{2}\right]$

k: $\tan^{-1}5x$ domain R

5. Apply $2Cos^{-1}x=Cos^{-1}(2x^2-1)$ formula

6. $-\frac{\pi}{7}+\frac{\pi}{3}-\frac{2\pi}{7}+\left(\frac{\pi}{2}-2\right)=\frac{17\pi}{42}-2$

7. Put $x=Cos\theta\Rightarrow\sqrt{\frac{1-x}{2}}=Sin\theta$

8. Verify with option

9. Apply $\sin^{-1}x+\sin^{-1}y$ formula

10. Apply $Tan^{-1}x+Tan^{-1}y$ formula

11. Put $x=Tan\theta$

12. Apply $Tan^{-1}x+Tan^{-1}y$ formula

13. $Tan[Tan^{-1}(\frac{1}{a+b})+Tan^{-1}(\frac{(a+b)-a}{1+(a+b)a})]$

$=Tan[Tan^{-1}(\frac{1}{a+b})+Tan^{-1}(a+b)-Tan^{-1}a]$

$=Tan[\frac{\pi}{2}-Tan^{-1}(a)]=\frac{1}{a}$

14. Applying $Tan^{-1}x+Tan^{-1}y$ formula and substitude $c^2=a^2+b^2$

15. By verification x=2 satisfied

16. $x(x+1)\geq 0$ and $x^2+x+1\geq 0$

But $\sin^{-1}(\sqrt{x})$ domain [0, 1]

$\therefore$ x(x+1) = 0 $\Rightarrow$ x = 0 or –1

17. $x=\frac{3\pi}{4}, \quad y=Cos\frac{\alpha}{2}$ put

$Cos^{-1}\left(\frac{1}{8}\right)=\alpha, \quad y=\sqrt{\frac{1+Cos\alpha}{2}}=\frac{3}{4}$

18. use $Tan^{-1}x+Tan^{-1}y+Tan^{-1}z=Tan^{-1}\left(\frac{\sum x-\Pi x}{1-\sum xy}\right)$

(or) Put a=b=c=1 verify option

19. $2Tan^{-1}y=Tan^{-1}x+Tan^{-1}z \Rightarrow y^2=xz$

$\therefore x, y, z$ are in G.P as well as in A.P
$x=y=z$

20. $1 \le Sin^{-1}Cos^{-1}Sin^{-1}Tan^{-1}x \le \frac{\pi}{2}$

$\Rightarrow Sin1 \le Cos^{-1}Sin^{-1}Tan^{-1}x \le 1$
$\Rightarrow CosSin1 \ge Sin^{-1}Tan^{-1}x \ge Cos1$
$\Rightarrow SinCosSin1 \ge Tan^{-1}x \ge SinCos1$
$\Rightarrow TanSinCosSin1 \ge x \ge TanSinCos1$

21. Given $3x-x^2 \ge 2 \Rightarrow x^2-3x+2 \le 0 \Rightarrow x=1(or)2$
At x=2 sine is not defined
So $Sin^{-1}(1)+Sin^{-1}(1)+.........+10$ times

$=\frac{\pi}{2}+\frac{\pi}{2}+.....+10$ times $=5\pi$

22. $Tan^{-1}x-Tan^{-1}y=0 \Rightarrow x=y$

$Cos^{-1}x+Cos^{-1}y=\frac{\pi}{2} \Rightarrow x=\frac{1}{\sqrt{2}}$

23. Apply $Cot^{-1}x=Cos^{-1}(\frac{x}{\sqrt{1+x^2}})$ and

$Cot^{-1}x=Sin^{-1}(\frac{1}{\sqrt{1+x^2}})$ formulas

24. $x^2-2x+2=(x-1)^2+1 \ge 1 \therefore x=1$

$a+\frac{\pi}{2}=0 \Rightarrow a=-\frac{\pi}{2}$

25. $(Sin^{-1}x)^2+(Cos^{-1}x)^2=(Sin^{-1}x)^2+(\frac{\pi}{2}-Sin^{-1}x)^2$

$=2(Sin^{-1}x)^2-\pi Sin^{-1}x+\frac{\pi^2}{4}$

$\therefore$ Its minimum value is $\frac{4.2.\frac{\pi^2}{4}-\pi^2}{8}=\frac{\pi^2}{8}$

26. $(Tan^{-1}x+Cot^{-1}x)^2-2Tan^{-1}x(\frac{\pi}{2}-Tan^{-1}x)=\frac{5\pi^2}{8}$

$2(Tan^{-1}x)^2-\pi Tan^{-1}x-\frac{3\pi^2}{8}=0$

$\Rightarrow Tan^{-1}x=-\frac{\pi}{4},\frac{3\pi}{4}$

$\Rightarrow Tan^{-1}x=-\frac{\pi}{4} \Rightarrow x=-1$

27. $Sin^{-1}(x^2-6x+12)=\frac{\pi}{3}$

$x^2-6x+12=(x-3)^2+3 \ge 3 \quad \forall x \in R$

28. $If -1 \le x < \frac{1}{\sqrt{2}} \Rightarrow Cos^{-1}x > Sin^{-1}x$

29. $2\ Tan^{-1}x > \frac{\pi}{2}$.

30. $Sin^{-1}(Sin12)+Cos^{-1}(Cos12)=12-4\pi+4\pi-12=0$

31. Apply $Cos^{-1}x-Cos^{-1}y=Cos^{-1}(xy+\sqrt{1-x^2}\sqrt{1-y^2})$

32. $\log^{x}{}_{2} \ge 0 \Rightarrow 1$, for

$x \ge 1, Sin^{-1}(\frac{2x}{1+x^2})=\pi-2Tan^{-1}x$

33. $Sin^{-1}x+Sin^{-1}y+Sin^{-1}z=\frac{3\pi}{2} \Rightarrow x=y=z=1$

34. common domain of P,Q,R is {-1,1}

min of P is at $x=-1 \Rightarrow P=-\frac{3\pi}{2}$

min of Q is at $x=-1 \Rightarrow Q=\frac{\pi}{2}$

min of R is at $x=1 \Rightarrow R=\frac{-\pi}{2}$

35. $\sum_{n=1}^{n} \tan^{-1}\left(\frac{1}{1+(n-1)n}\right) = \tan^{-1}(n) - \tan^{-1}(1)$

LEVEL - II (H.W)

1. **Range of** $Sin^{-1}x - Cos^{-1}x$ **is**

1) $[\frac{-3\pi}{2}, \frac{\pi}{2}]$ 2) $[\frac{-5\pi}{3}, \frac{\pi}{3}]$

3) $[\frac{-3\pi}{2}, \pi]$ 4) $[0, \pi]$

2. **The ascending order of** $A = \text{Sin}^{-1}\left(\log_3{}^2\right)$, $B = \text{Cos}^{-1}\left(\log_3\left(\frac{1}{2}\right)\right)$, $C = \text{Tan}^{-1}\left(\log_{1/3} 2\right)$ **is**

1) C,B,A 2) B,A,C 3) C, A,B 4) B,C,A

3. **The domain of** $Tan^{-1}(5x)$

1) R 2) $(0, \pi)$ 3) $[0, \pi]$ 4) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

4. **The decreasing order of ,**

$A = \left(\text{Sin}^{-1}\left(\frac{1-\sqrt{3}}{2\sqrt{2}}\right)\right)$ $B = \text{Cos}^{-1}\left(\frac{-1}{\sqrt{2}}\right)$,

$C = \text{Tan}^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$ **is**

1)B,A,C 2)B,C,A 3)C,A,B 4)C,B,A

5. **The value of** $Sin^{-1}(Cos(\frac{33\pi}{5}))$ **is**

1) $\frac{3\pi}{5}$ 2) $\frac{7\pi}{5}$ 3) $\frac{\pi}{10}$ 4) $\frac{-\pi}{10}$

6. $\text{Cos}^{-1}\left(\cos\frac{10\pi}{7}\right) + \text{Sin}^{-1}\left(\sin\frac{35\pi}{11}\right) + \text{Tan}^{-1}\left(\tan\frac{24\pi}{13}\right) + Cot^{-1}\left(\cot\frac{26\pi}{5}\right)$ **is**

1) $\frac{4\pi}{7} + \frac{\pi}{55} + \frac{2\pi}{13}$ 2) $\frac{4\pi}{7} - \frac{\pi}{55} - \frac{2\pi}{13}$

3) $\frac{4\pi}{7} - \frac{\pi}{55} + \frac{2\pi}{13}$ 4) $-\frac{4\pi}{7} + \frac{\pi}{55} + \frac{2\pi}{13}$

7. **The value of tan** $\left(\frac{1}{2}\text{Cos}^{-1}\left(\frac{\sqrt{5}}{3}\right)\right)$ **is**

1) $\frac{3+\sqrt{5}}{2}$ 2) $3+\sqrt{5}$

3) $\frac{1}{2}\left(3-\sqrt{5}\right)$ 4) $2-\sqrt{3}$

8. **If** $Tan^{-1}\frac{5}{12} + Tan^{-1}\frac{24}{25} = Cos^{-1}x$ **then x =**

1) $\frac{-31}{325}$ 2) $\frac{-33}{325}$ 3) $\frac{-36}{325}$ 4) $\frac{-39}{325}$

9. $6Sin^{-1}(x^2 - 6x + 8.5) = \pi$ **, then x =**

1) 1 2) 2 3) 3 4) 0

10. **If Tan^{-1} (x+1) + Tan^{-1}(x -1) = Tan^{-1}** $\frac{8}{31}$ **then x =**

1) 1 2) 1/2 3) -1/2 4) 1/4

11. **If Cot^{-1}** $\left(\frac{1}{x+1}\right)$ **+ Cot^{-1}** $\left(\frac{1}{x-1}\right)$ **=** Tan^{-1} 3x - Tan^{-1} x **then** $x =$

1) ±1/2 2) -1, ± 1/3 3) 2, ±1 4) -1,±1/2

12. **Tan^{-1}** $\frac{5}{6}$ + $\frac{1}{2}$ **Tan^{-1}** $\frac{11}{60}$ =

1) π 2) $\pi/2$ 3) $\pi/4$ 4) $3\pi/4$

13. $Tan\left(\frac{\pi}{4} + \frac{1}{2}Cos^{-1}\frac{a}{b}\right) + Tan\left(\frac{\pi}{4} - \frac{1}{2}Cos^{-1}\frac{a}{b}\right)$

1) b/a 2) a/b 3) 2a/b 4) 2b/a

14. $Cos^{-1}\left\{\frac{1}{\sqrt{2}}\left(Cos\frac{9\pi}{10} - Sin\frac{9\pi}{10}\right)\right\} =$

1) $\frac{3\pi}{20}$ 2) $\frac{7\pi}{10}$ 3) $\frac{7\pi}{20}$ 4) $\frac{17\pi}{20}$

15. **If** $Tan^{-1}(\frac{x-1}{x-2}) + Tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ **then** $x =$

1) $\frac{1}{\sqrt{2}}$ 2) $\pm\frac{1}{\sqrt{2}}$ 3) $\pm\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

16. **The number of solutions of the equation** $2\left(Sin^{-1}x\right)^2 - 5Sin^{-1}x + 2 = 0$ **is**

1) 0 2) 1 3) 2 4) 3

17. **If** $\frac{1}{\sqrt{2}} < x < 1$ **then**

$Cos^{-1}x + Cos^{-1}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{2}}\right) =$

1) $2Cos^{-1}x - \frac{\pi}{4}$ 2) $2Cos^{-1}x$ 3) $\frac{\pi}{4}$ 4) 0

18. **Value of** $\mathrm{Tan}^{-1}2 + \mathrm{Tan}^{-1}3 + \mathrm{Tan}^{-1}4 =$

1) $\pi - \mathrm{Tan}^{-1}\frac{3}{5}$ 2) $\pi + \mathrm{Tan}^{-1}\frac{5}{3}$

3) $\pi - \mathrm{Tan}^{-1}\frac{5}{3}$ 4) $\pi + \mathrm{Tan}^{-1}\frac{3}{5}$

19. **The function** $f(x) = \mathrm{Tan}^{-1}(\sin x + \cos x)$ **is an increasing function in**

1) $(\frac{\pi}{4}, \frac{\pi}{2})$ 2) $(-\frac{\pi}{2}, \frac{\pi}{4})$ 3) $(0, \frac{\pi}{2})$ 4) $(-\frac{\pi}{2}, \frac{\pi}{2})$

20. **If** $\left[Cot^{-1}x\right] + \left[\mathrm{Cos}^{-1}x\right] = 0$ **where** x **is non-negative real number and** $[\bullet]$ **denotes the greatest integer function, then complete set of values of** x **is**

1) $(\cos 1,\ 1]$ 2) $(\cos 1,\ \cot 1)$

3) $(\cot 1\ \ 1]$ 4) $(0,\ \cos 1)$

21. **For the equation** $Cos^{-1}x + \mathrm{Cos}^{-1}2x + \pi = 0$ **the number of real solution is**

1) 0 2) 1 3) 2 4) 3

22. **If** $\alpha = 2\mathrm{Tan}^{-1}\left(\sqrt{2}-1\right)$,

$\beta = 3\mathrm{Sin}^{-1}\left(\frac{1}{\sqrt{2}}\right) + \mathrm{Sin}^{-1}(-\frac{1}{2})$ **and**

$\gamma = \mathrm{Cos}^{-1}\frac{1}{3}$ **then**

1) $\alpha < \beta < \gamma$ 2) $\alpha < \gamma < \beta$

3) $\beta < \gamma < \alpha$ 4) $\gamma < \beta < \alpha$

23. **The value of**

$\mathrm{Sin}^{-1}\left(\cot\left(\mathrm{Sin}^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \mathrm{Cos}^{-1}\frac{\sqrt{12}}{4} + \mathrm{Sec}^{-1}\sqrt{2}\right)\right)$ **is**

1) 0 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) π

24. **The equation** $2\mathrm{Cos}^{-1}x + \mathrm{Sin}^{-1}x = \frac{11\pi}{6}$ **has**

1) No solution 2) One solution

3) Two solutions 3) Three solutions

25. **The smallest and the largest values of**

$\mathrm{Tan}^{-1}\left(\frac{1-x}{1+x}\right), 0 \le x \le 1$ **are**

1) $0, \pi$ 2) $0, \frac{\pi}{4}$ 3) $-\frac{\pi}{4}, \frac{\pi}{4}$ 4) $\frac{\pi}{4}, \frac{\pi}{2}$

26. **If** α, β, γ **are the roots of the equation** $x^3 + mx^2 + 3x + m = 0$**, then the general value of** $\mathrm{Tan}^{-1}\alpha + \mathrm{Tan}^{-1}\beta + \mathrm{Tan}^{-1}\gamma$ **is**

1) $(2n+1)\frac{\pi}{2}$ 2) $n\pi$ 3) $\frac{n\pi}{2}$

4) dependent upon the value of m

27. **For** $0 \le Cos^{-1}x \le \pi$ **and** $-\frac{\pi}{2} \le Sin^{-1}x \le \frac{\pi}{2}$

then value of $Cos\left(Sin^{-1}x + 2Cos^{-1}x\right)$ **at** $x = \frac{1}{5}$ **is**

1) $\frac{-2\sqrt{6}}{5}$ 2) $\frac{-\sqrt{6}}{5}$ 3) $\frac{2\sqrt{6}}{5}$ 4) $\frac{\sqrt{6}}{5}$

28. **If** $'a'$ **is twice the tangent of the arithmetic mean of** $\mathrm{Sin}^{-1}x$ **and** $\mathrm{Cos}^{-1}x$ **,** $'b'$ **is the geometric mean of** $\tan x$ **and** $\cot x$**. Then** $x^2 - ax + b = 0 \Rightarrow x =$

1) 2 2) 3 3) 1 4) 0

29. **If x>1 then** $2\mathrm{Tan}^{-1}x$ **cannot be equal to**

1) $\mathrm{Cos}^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ 2) $\pi - \mathrm{Sin}^{-1}\left(\frac{2x}{1+x^2}\right)$

3) $\pi + Tan^{-1}\left(\frac{2x}{1-x^2}\right)$ 4) $\pi - \mathrm{Cos}^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

30. $Cos^{-1}(Cos12) - Sin^{-1}(Sin12) =$

1) 0 2) π 3) $8\pi + 24$ 4) $8\pi - 24$

31. **If** $Cos^{-1}\frac{x}{a} + Cos^{-1}\frac{y}{b} = \frac{5\pi}{12}$ **and**

$Sin^{-1}\frac{x}{a} - Sin^{-1}\frac{y}{b} = \frac{\pi}{12}$ **then** $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$

1) 1 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) $\frac{5}{4}$

32. $2Cos^{-1}x = Sin^{-1}(2x\sqrt{1-x^2})$ **is valid for**

1) $-1 \le x \le 1$ 2) $0 \le x \le 1$

3) $0 \le x \le \frac{1}{\sqrt{2}}$ 4) $\frac{1}{\sqrt{2}} \le x \le 1$

33. **If** $Tan^{-1}x + Tan^{-1}y + Tan^{-1}z = \frac{\pi}{2}$ **and** $(x-y)^2 + (y-z)^2 + (z-x)^2 = 0$ **then** $x^2 + y^2 + z^2 =$

1) 0 2) 4 3) 1 4) 2

34. **The ascending order of** $A = Cos(Sin^{-1}\frac{12}{13})$ $B = Sin(2Tan^{-1}\frac{3}{4})$ $C = Tan(\frac{1}{2}Cos^{-1}\frac{1}{3})$ **is**

1) A, B, C 2) A, C, B 3) B, C, A 4) B, A, C

35. $\sum_{m=1}^{n} \text{Tan}^{-1}\left(\frac{2m}{m^4+m^2+2}\right) =$

1. $\text{Tan}^{-1}\left(\frac{n^2+n}{n^2+n+2}\right)$ 2. $\text{Tan}^{-1}\left(\frac{n^2-n}{n^2-n+2}\right)$

3. $\text{Tan}^{-1}\left(\frac{n^2+n+2}{n^2+n}\right)$ 4. $\frac{\pi}{4}$

LEVEL - II (H.W)-KEY

01) 1	02)3	03) 1	04) 2	05) 4
06) 1	07) 3	08) 3	09) 2	10) 4
11) 1	12) 3	13) 4	14) 4	15) 2
16) 2	17) 3	18) 4	19) 2	20) 3
21) 1	22) 2	23) 1	24) 1	25) 2
26) 2	27) 1	28) 3	29) 4	30) 4
31) 4	32) 4	33) 3	34) 2	35) 1

LEVEL - II (H.W)-HINTS

1. Let $y = Sin^{-1}x - Cos^{-1}x = \frac{\pi}{2} - 2Cos^{-1}x$

$0 \le Cos^{-1}x \le \pi \Rightarrow -\frac{3\pi}{2} \le y \le \frac{\pi}{2}$

2. $0 < A < 90^0,\ B > 90^0, C < 0$

3. Domain of $Tan^{-1}x$ is R

4. $A < 0, B > 90^0, 0 < C < 90^0$

5. $\frac{33\pi}{5} = 6\pi + \frac{3\pi}{5} \in Q1$

$Sin^{-1}[Cos(\frac{33\pi}{5})] = Sin^{-1}[Cos(\frac{3\pi}{5})]$

6. $\cos^{-1}\cos\left(\frac{10\pi}{7}\right) =$

$\cos^{-1}\cos\left(2\pi - \frac{10\pi}{7}\right) = \cos^{-1}\cos\left(\frac{4\pi}{7}\right) = \left(\frac{4\pi}{7}\right)$

$\sin^{-1}\sin\left(\frac{35\pi}{11}\right) = \sin^{-1}\sin\left(2\pi + \frac{13\pi}{11}\right) +$

$\sin^{-1}\sin\left(\frac{13\pi}{11}\right) = \sin^{-1}\sin\left(\pi + \frac{2\pi}{11}\right)$

$= \sin^{-1}\sin\left(-\frac{2\pi}{11}\right) = -\frac{2\pi}{11}$

$\tan^{-1}\tan\left(\frac{24\pi}{13}\right) = \tan^{-1}\tan\left(\frac{2\pi}{13} - 2\pi\right) =$

$\tan^{-1}\tan\left(\frac{2\pi}{13}\right) = \frac{2\pi}{13}$]

$\cot^{-1}\left(\cot\frac{26\pi}{5}\right) = \cot^{-1}\left(\cot\left(5\pi + \frac{\pi}{5}\right)\right) = \frac{\pi}{5}$

7. Put $Cos^{-1}\frac{\sqrt{5}}{3} = x$

8. $Sin^{-1}x = Tan^{-1}\frac{x}{\sqrt{1-x^2}} \Rightarrow Sin^{-1}\frac{24}{25} = Tan^{-1}\frac{24}{7}$

Apply $Tan^{-1}x + Tan^{-1}y = x + Tan^{-1}\left(\frac{x+y}{1-xy}\right)$, $xy > 1$ formula

9. $6\sin^{-1}\left(x^2 - 6x + \frac{17}{2}\right) = \pi$

$\sin^{-1}\left(x^2 - 6x + \frac{17}{2}\right) = \frac{\pi}{6}$

$x^2 - 6x + \frac{17}{2} = \frac{1}{2}$ solve the equation. It has two values.

10. Apply $Tan^{-1}x + Tan^{-1}y$ formula

11. $Tan^{-1}(x+1) + Tan^{-1}(x-1) = Tan^{-1}3x - Tanx$

Apply $Tan^{-1}x + Tan^{-1}y$ and

$Tan^{-1}x - Tan^{-1}y$ formula

12. $\frac{1}{2}[2Tan^{-1}\frac{5}{6} + Tan^{-1}\frac{11}{60}] = \frac{1}{2}[Tan^{-1}\frac{60}{11} + Tan^{-1}\frac{11}{60}] = \frac{1}{2}\frac{\pi}{2} = \frac{\pi}{4}$

13. Put $\frac{1}{2}Cos^{-1}\frac{a}{b} = \theta \Rightarrow Cos2\theta = \frac{a}{b}$

Apply $Tan(\frac{\pi}{4} + \theta) + Tan(\frac{\pi}{4} - \theta) = \frac{2}{Cos2\theta}$

14. $Cos^{-1}[Cos(\frac{9\pi}{10} + \frac{\pi}{4})] = Cos^{-1}[Cos(\frac{23\pi}{20})]$

$= Cos^{-1}[Cos(2\pi - \frac{17\pi}{20})]$

$= Cos^{-1}[Cos(\frac{17\pi}{20})] = \frac{17\pi}{20}$

15. Apply $Tan^{-1}x + Tan^{-1}y$ formula

16. Put $\sin^{-1}x = a$

17. $Cos^{-1}x + Cos^{-1}(x.\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\sqrt{1-x^2})$

$= Cos^{-1}x + Cos^{-1}(\frac{1}{\sqrt{2}}) - Cos^{-1}x = \frac{\pi}{4}$

18. Apply $Tan^{-1}x + Tan^{-1}y = \pi + Tan^{-1}(\frac{x+y}{1-xy}), xy > 1$

And $Tan^{-1}x - Tan^{-1}y$ formula

19. $f(x) = Tan^{-1}(Sinx + Cosx)$ find $f^1(x)$

$f'(x) > 0$ if $-\frac{\pi}{2} < x + \frac{\pi}{4} < \frac{\pi}{2}$ then

$-\frac{3\pi}{4} < x < \frac{\pi}{4}$ Hence $f(x)$ is increasing

when $x \in (-\frac{\pi}{2}, \frac{\pi}{4})$

20. $0 < Cot^{-1}x < \pi$ and $0 \le Cos^{-1}x \le \pi$

$Cot\ \ 1 < x < \infty$ and $Cos\ \ 1 \le x \le 1$

$x \in (Cot1, 1]$ $\quad \because Cos\ \ 1 < Cot1$

21. We have $\text{Cos}^{-1}x + \text{Cos}^{-1}2x = -\pi$ which is not possible as $\text{Cos}^{-1}x$ *and* $\text{Cos}^{-1}2x$ never take negative values

22. $\alpha = 2Tan^{-1}(\sqrt{2} - 1) = \frac{\pi}{4}$, $\beta = \frac{7\pi}{12}$,

$\gamma = Cos^{-1}(\frac{1}{3}) > \frac{\pi}{4}$

23. $Sin^{-1}[Cot(\frac{\pi}{12} + \frac{\pi}{6} + \frac{\pi}{4})] = Sin^{-1}[Cot(\frac{\pi}{2})]$

$= Sin^{-1}(0) = 0$

$\because Sin^{-1}x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

24. $Cos^{-1}x + (Cos^{-1}x + Sin^{-1}x) = \frac{11\pi}{6} \Rightarrow Cos^{-1}x + \frac{\pi}{2} = \frac{11\pi}{6}$

$\Rightarrow Cos^{-1}x = \frac{4\pi}{3}$ which is not possible as

$Cos^{-1}x \in [0, \pi]$

25. We have $Tan^{-1}(\frac{1-x}{1+x}) = \frac{\pi}{4} - Tan^{-1}x$

Since

$0 \le x \le 1 \Rightarrow 0 \le Tan^{-1}x \le \frac{\pi}{4} \Rightarrow 0 \le Tan^{-1}(\frac{1-x}{1+x}) \le \frac{\pi}{4}$

26. Let $\alpha = Tan\theta_1$, $\beta = Tan\theta_2$, $\gamma = Tan\theta_3$

$Tan(\theta_1 + \theta_2 + \theta_3) \Rightarrow \theta_1 + \theta_2 + \theta_3 = n\pi,\ n \in z$

27. $Cos(\frac{\pi}{2} + Cos^{-1}x) = -\sqrt{1-x^2}$

28. $a = 2Tan(\frac{Sin^{-1}x + Cos^{-1}x}{2}) = 2(1) = 2$

$b = \sqrt{Tanx.Cotx} = 1$

$\therefore x^2 - ax + b = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow x = 1$

29. if x>1 then $2\tan^{-1}x > \frac{\pi}{2}$ by verification

$\pi-\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is not equal to $2\tan^{-1}x$.

30. $(4\pi-12)-(12-4\pi)=8\pi-24$

31. $Cos^{-1}\frac{x}{a}+Cos^{-1}\frac{y}{b}=\frac{5\pi}{2}\to(1)$ and

$Cos^{-1}\frac{y}{b}-Cos^{-1}\frac{x}{a}=\frac{\pi}{2}\to(2)$

$(1)+(2)\Rightarrow Cos^{-1}\frac{y}{b}=\frac{\pi}{4}\Rightarrow\frac{y}{b}=\frac{1}{\sqrt{2}}$

$(1)-(2)\Rightarrow Cos^{-1}\frac{x}{a}=\frac{\pi}{6}\Rightarrow\frac{x}{a}=\frac{\sqrt{3}}{2}$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{5}{4}$

32. Let $x=Cosy$ where $0\le y\le\pi$, $|x|\le1$

$2Cos^{-1}x=Sin^{-1}(2x\sqrt{1-x^2})\to(1)$

$\Rightarrow 2Cos^{-1}(Cosy)=Sin^{-1}(2Cosy\sqrt{1-Cos^2y})$

$\Rightarrow 2Cos^{-1}(Cosy)=Sin^{-1}(Sin2y)$

$Sin^{-1}(Sin2y)=2y$ for $-\frac{\pi}{4}\le y\le\frac{\pi}{4}$

And $2Cos^{-1}(Cosy)=2y$ for $0\le y\le\pi$

Thus equation (1) holds only when

$$y\in[0,\frac{\pi}{4}]\Rightarrow x\in[\frac{1}{\sqrt{2}},1]$$

33. Put $x=y=z=\frac{1}{\sqrt{3}}$

$x^2+y^2+z^2=\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=1$

34. $A=Cos(Sin^{-1}\frac{12}{13})=\frac{5}{13}$

$\left(Sin^{-1}x=Cos^{-1}\sqrt{1-x^2}\right)$

$B=Sin(2Tan^{-1}\frac{3}{4})=\frac{24}{25}\left(\because 2Tan^{-1}x=Sin^{-1}(\frac{2x}{1+x^2})\right)$

$C=Tan(\frac{1}{2}Cos^{-1}\frac{1}{3})$ put $Cos^{-1}\frac{1}{3}=\theta$

then $C=\frac{1}{\sqrt{2}}$

35. $\sum_{m=1}^{n}\tan^{-1}(\frac{2m}{m^4+m^2+2})$

$=\sum_{m=1}^{n}\tan^{-1}(\frac{(m^2+m+1)-(m^2-m+1)}{1+(m^2+m+1)(m^2-m+1)})$

$=\sum_{m=1}^{n}[\tan^{-1}(m^2+m+1)-\tan^{-1}((m^2-m+1)]$

LEVEL - III

1. $\frac{Cos^{-1}(41/49)}{Sin^{-1}(2/7)}=$

1) 4 2) 3 3) 2 4) 1

2. If x takes negative permissible value, then

$\mathrm{Sin}^{-1}x=$

1) $\mathrm{Cos}^{-1}\sqrt{1-x^2}$ 2) $-\mathrm{Cos}^{-1}\sqrt{1-x^2}$

3) $\mathrm{Cos}^{-1}\sqrt{x^2-1}$ 4) $\pi-\mathrm{Cos}^{-1}\sqrt{1-x^2}$

3. $\mathrm{Cos}^{-1}\sqrt{\frac{a-x}{a-b}}=\mathrm{Sin}^{-1}\sqrt{\frac{x-b}{a-b}}$ $(a\ne b)$ **is possible if**

1) $a\ge x\ge b$ or $a\le x\le b$
2) $a=x=b$
3) $a>b$ and x takes any value
4) $a<b$ and x takes any value

4. The value of $\mathrm{Sin}^{-1}\left(\cos\left\{\mathrm{Cos}^{-1}(\cos x)+\mathrm{Sin}^{-1}(\sin x)\right\}\right)$

where $x\in\left(\frac{\pi}{2},\pi\right)$ **is**

1) $\frac{\pi}{2}$ 2) $-\pi$ 3) π 4) $-\frac{\pi}{2}$

5. If $x=\sin(2Tan^{-1}2)$ **and** $y=sin\left[\frac{1}{2}Tan^{-1}\frac{4}{3}\right]$

then

1) $x = y$ 2) $x > y$ 3) $\frac{2}{\sqrt{5}}$ 4) $x < y$

6. **$Sin^{-1}(\sin 3) + Cos^{-1}(\cos 7) - Tan^{-1}(\tan 5) =$**

1) $\pi - 1$ 2) π 3) $3\pi - 1$ 4) $2\pi - 1$

7. **The number of solutions of**

$$\text{Sin}^{-1}\left(1+b+b^2+....\infty\right)+\text{Cos}^{-1}\left(a-\frac{a^2}{3}+\frac{a^3}{9}+...\infty\right)=\frac{\pi}{2}$$

1) 1 2) 2 3) 3 4) ∞

8. **If $y = (Sin^{-1}x)^3 + (Cos^{-1}x)^3$ then**

1) $\min y = \frac{\pi^3}{8}$ 2) $\min y = \frac{\pi^3}{32}$

3) $\max y = \frac{\pi^3}{8}$ 4) $\max y = \frac{7\pi^3}{32}$

9. $Sin^{-1}(\frac{1}{\sqrt{2}})+Sin^{-1}(\frac{\sqrt{2}-1}{\sqrt{6}})+....+Sin^{-1}(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}})+....\infty=$

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{2}$

10. $\text{Cos}^{-1}\left\{\frac{1}{2}x^2+\sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right\}=\text{Cos}^{-1}\frac{x}{2}-\text{Cos}^{-1}x$

holds for :

1) $|x| \le 1$ 2) $x \in R$

3) $0 \le x \le 1$ 4) $-1 \le x \le 0$

11. **The set of values of x from which the identity** $\text{Cos}^{-1}x+\text{Cos}^{-1}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right)=\frac{\pi}{3}$ **holds good is**

1) $[0,1]$ 2) $[0,\frac{1}{2}]$ 3) $[\frac{1}{2},1]$ 4) $\{-1,0,1\}$

12. **If A =** $Tan^{-1}\left(\frac{x\sqrt{3}}{2k-x}\right)$ **and B =** $Tan^{-1}\left(\frac{2x-k}{k\sqrt{3}}\right)$ **then A-B =**

1) 0^0 2) $\pi/6$ 3) $\pi/4$ 4) $\pi/3$

13. **The number of solutions of the equation** $Tan^{-1}(x-1) + Tan^{-1}(x)+Tan^{-1}(x+1)=Tan^{-1}(3x)$ **is**

1) 1 2) 2 3) 3 4) 4

14. $Cot^{-1}\left(\sqrt{\cos\alpha}\right)-\text{Tan}^{-1}\left(\sqrt{\cos\alpha}\right)=x\ge 0$, **then sin x =** **(AIE-2002)**

1) $tan^2\frac{\alpha}{2}$ 2) $\cot^2\left(\frac{\alpha}{2}\right)$ 3) $\tan\alpha$ 4) $\cot\frac{\alpha}{2}$

15. **If a_1, a_2, a_3 .... a_n are in A.P. with common difference 'd', then**

$$tan\left[Tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+Tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+......+Tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]=$$

1) $\frac{(n-1)d}{a_1+a_n}$ 2) $\frac{(n-1)d}{1+a_1a_n}$

3) $\frac{nd}{1+a_1a_n}$ 4) $\frac{a_n-a_1}{a_n+a_1}$

16. $Tan^{-1}(\frac{c_1x-y}{c_1y+x})+Tan^{-1}(\frac{c_2-c_1}{1+c_2c_1})+Tan^{-1}(\frac{c_3-c_2}{1+c_3c_2})+....+Tan^{-1}(\frac{1}{c_n})=$

1) $Tan^{-1}\left(\frac{2x}{y}\right)$ 2) $Tan^{-1}(xy)$

3) $Tan^{-1}\left(\frac{x}{y}\right)$ 4) $Tan^{-1}\left(\frac{y}{x}\right)$

17. $\text{Tan}^{-1}\left(\frac{1}{3}\right)+\text{Tan}^{-1}\left(\frac{2}{9}\right)+\text{Tan}^{-1}\left(\frac{4}{33}\right)+......\infty$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) 2π

18. **Value of** $\text{Sec}^{-1}\left(\frac{1}{1-2x^2}\right)+4\text{Cos}^{-1}\sqrt{\frac{1+x}{2}}=$

1) $2\text{Tan}^{-1}x$ 2) $\text{Tan}^{-1}\left(\frac{\sqrt{1-x^2}}{1-x}\right)$

3) $Cot^{-1}\left(\frac{\sqrt{1-x^2}}{1-x}\right)$ 4) constant for all x

19. **The solution set of the equation** $\text{Tan}^{-1}x-Cot^{-1}x=\text{Cos}^{-1}(2-x)$ **is :**

1) $[0,1]$ 2) $[-1,1]$ 3) $[1,3)$ 4) $(1,3)$

20. The number of solutions of the equation

$3\,\text{Cos}^{-1}x - \pi x - \frac{\pi}{2} = 0$

1) 0 2) 1 3) 2 4) infinitely many

21. Number of real solutions of the equation

$\sqrt{1+\cos 2x} = \sqrt{2}\,\text{Sin}^{-1}(\sin x)$ **where**

$-\pi \le x \le \pi$

1) 0 2) 1 3) 2 4) 4

22. If the equation

$\text{Sin}^{-1}\left(x^2+x+1\right)+\text{Cos}^{-1}(\lambda x+1)=\frac{\pi}{2}$

has exactly two solutions, then λ can have the integral value.

1) -1 2) 0 3) 1 4) 2

23. If x_1, x_2, x_3, x_4 are roots of the equation

$x^4 - x^3\sin 2\beta + x^2\cos 2\beta - x\cos\beta - \sin\beta = 0$

then $\text{Tan}^{-1}x_1 + \text{Tan}^{-1}x_2 + \text{Tan}^{-1}x_3 + \text{Tan}^{-1}x_4 =$

1) β 2) $\frac{\pi}{2}-\beta$ 3) $\pi-\beta$ 4) $-\beta$

24. If $Cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}, n\in N$ **then the maximum value of 'n' is**

1) 6 2) 7 3) 5 4) 10

25. If $\left(Cot^{-1}x\right)^2 - 7\left(Cot^{-1}x\right)+10>0$**, then x lies in the interval**

1) $(\cot 5, \cot 2)$ 2) $(-\infty, \cot 5)\cup(\cot 2, \infty)$

3) $(-\infty, \cot 5)$ 4) $(\cot 2, \infty)$

26. The value of x for which

$\text{Cos}^{-1}(\cos 4) > 3x^2 - 4x$ **is**

1) $\left(0, \frac{2+\sqrt{6\pi-8}}{3}\right)$ 2) $\left(\frac{2+\sqrt{6\pi-8}}{3}, 0\right)$

3) $(-2, 2)$ 4) $\left(\frac{2-\sqrt{6\pi-8}}{3}, \frac{2+\sqrt{6\pi-8}}{3}\right)$

27. The least integral value of k for which $(k-2)x^2 + 8x + k + 4 > \text{Sin}^{-1}(\sin 12) + \text{Cos}^{-1}(\cos 12)$ for all $x \in R$, is

1) -7 2) -5 3) -3 4) 5

28. Point P(x, y) satisfying the equation

$\text{Sin}^{-1}x + \text{Cos}^{-1}y + \text{Cos}^{-1}(2xy) = \frac{\pi}{2}$ **lies on**

1) the bisector of the first and third quadrant

2) bisector of the second and fourth quadrant.

3) the rectangle formed by the lines $x = \pm 1$ and $y = \pm 1$.

4) a unit circle with centre at the origin.

29. If α is the only real root of the equation $x^3 + bx^2 + cx + 1 = 0 (b < c)$ **then the value of**

$\text{Tan}^{-1}\alpha + \text{Tan}^{-1}\left(\frac{1}{\alpha}\right) =$

1) $\frac{\pi}{2}$ 2) $-\frac{\pi}{2}$ 3) 0 4) π

30. $\frac{\alpha^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\text{Tan}^{-1}\frac{\alpha}{\beta}\right) + \frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\text{Tan}^{-1}\left(\frac{\beta}{\alpha}\right)\right) =$

1) $(\alpha-\beta)(\alpha^2+\beta^2)$ 2) $(\alpha+\beta)(\alpha^2-\beta^2)$

3) $(\alpha+\beta)(\alpha^2+\beta^2)$ 4) 0

31. $x = n\pi - \text{Tan}^{-1}3$ **is a solution of the equation** $12\tan 2x + \frac{\sqrt{10}}{\cos x} + 1 = 0$ **if**

1) n is any integer

2) n is an even integer

3) n is a positive integer

4) n is an odd integer

LEVEL - III-KEY

01) 3	02) 2	03) 1	04) 4	05) 2	06) 1
07) 4	08) 2	09) 2	10) 3	11) 3	12) 2
13) 3	14) 1	15) 2	16) 3	17) 1	18) 4
19) 3	20) 2	21) 3	22) 2	23) 2	24) 3
25) 4	26) 4	27) 4	28) 4	29) 2	30) 3
31) 4					

LEVEL - III-HINTS

1. $\cos^{-1}\frac{41}{49} = 2\sin^{-1}\left(\frac{2}{7}\right)$

2. Let $\sin^{-1}x = y \Rightarrow x = \sin y$

since $-1 \le x \le 0$ therefore $-\frac{\pi}{2} \le \sin^{-1}x \le 0$

$\Rightarrow \frac{\pi}{2} \ge -y \ge 0$

$\Rightarrow (-y) = \cos^{-1}\sqrt{1-x^2} \Rightarrow y = -\cos^{-1}\sqrt{1-x^2}$

3. Take $\sqrt{\frac{a-x}{a-b}} = \sqrt{\frac{x-b}{a-b}} = \frac{1}{\sqrt{2}}$

4. $\sin^{-1}\left[\cos\{x+\pi-x\}\right] = -\pi/2$

5. Put $\tan^{-1}2$=A and $\tan^{-1}\frac{4}{3}$=B then verify

6. $\pi - 3 + 7 - 2\pi - (5-2\pi) = \pi - 1$

7. The given equation is valid if

$-1 \le 1+b+b^2+....\infty \le 1,\ and\ a - \frac{a^2}{3} + \frac{a^3}{9} - ....\infty \in [-1,1]$

also $1+b+b^2+....\infty = a - \frac{a^2}{3} + \frac{a^3}{9} - ....\infty$

8. By standed formula

$\frac{\pi^3}{32} \le \left(Sin^{-1}x\right)^3 + \left(Cos^{-1}x\right)^3 \le \frac{7\pi^3}{8}$

9. $T_n = Sin^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} = Tan^{-1}\frac{\sqrt{n}-\sqrt{n-1}}{1+\sqrt{n}\sqrt{n-1}}$

$= Tan^{-1}\sqrt{n} - Tan^{-1}\sqrt{n-1}$

$S_\infty = \left\{Tan^{-1}1 - Tan^{-1}0\right\} + \left\{Tan^{-1}\sqrt{2} - Tan^{-1}1\right\} + ... + Tan^{-1}\infty$

$= Tan^{-1}\infty - Tan^{-1}0 = \frac{\pi}{2}$

10. $Cos^{-1}\left(\frac{1}{2}x^2 + \sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right)$

$= \cos^{-1}\left(\frac{x}{2}x + \sqrt{1-x^2}\sqrt{1-\left(\frac{x}{2}\right)^2}\right)$

For $Cos^{-1}\left(\frac{1}{2}x^2 + \sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right)$

$= Cos^{-1}\frac{x}{2} - \cos^{-1}x$

L.H.S > 0, hence R.H.S>0

$\Rightarrow Cos^{-1}\frac{x}{2} - Cos^{-1}x > 0$

Since $Cos^{-1}x$ is decreasing function

$\frac{x}{2} \le x \Rightarrow \frac{x}{2} \ge 0 \Rightarrow x \in [0,1]$

11. by verification $x = \frac{1}{2}\ and\ 1$

12. Put x = k = 1

Apply $\tan^{-1}x - \tan^{-1}y$ formula

13. $\tan^{-1}(x-1) + \tan^{-1}(x+1) = \tan^{-1}(3x) - \tan^{-1}x$

apply the formula

14. $\cot^{-1}\sqrt{\cos\alpha} - \tan^{-1}\sqrt{\cos\alpha} = x$

$\Rightarrow \frac{\pi}{2} - 2\tan^{-1}\sqrt{\cos\alpha} = x$

$\Rightarrow \frac{\pi}{2} - \tan^{-1}\frac{2\sqrt{\cos\alpha}}{1-\cos\alpha} = x$

$\Rightarrow \cot^{-1}\left(\frac{2\sqrt{\cos\alpha}}{1-\cos\alpha}\right) = x \Rightarrow \cot x = \frac{2\sqrt{\cos\alpha}}{1-\cos\alpha}$

$\therefore \sin x = \frac{1-\cos\alpha}{1+\cos\alpha} = \tan^2\alpha/2$

15. Taking terms in numerator as

$d = a_2 - a_1 = a_3 - a_2$.......... after that apply

$\tan^{-1}x - \tan^{-1}y$ formula

16. $Tan^{-1}\left(\frac{\frac{x}{y}-\frac{1}{c_1}}{1+\frac{x}{y}.\frac{1}{c_1}}\right) + Tan^{-1}\left(\frac{\frac{1}{c_1}-\frac{1}{c_2}}{1+\frac{1}{c_1}.\frac{1}{c_2}}\right) + ... + Tan^{-1}\left(\frac{1}{c_n}\right)$

$\left\{Tan^{-1}\left(\frac{x}{y}\right) - Tan^{-1}\left(\frac{1}{c_1}\right)\right\} + \left\{Tan^{-1}\left(\frac{1}{c_1}\right) - Tan^{-1}\left(\frac{1}{c_2}\right)\right\} ..... + Tan^{-1}\left(\frac{1}{c_n}\right)$

$= Tan^{-1}\left(\frac{x}{y}\right)$

17. $T_n = \tan^{-1}\left(\frac{2^{n-1}}{1+2^{2n-1}}\right) = \tan^{-1}\left(\frac{2^n - 2^{n-1}}{1+2^n.2^{n-1}}\right)$

$= \tan^{-1} 2^{n} - \tan^{-1} 2^{n-1}$

$S_{\infty} = \tan^{-1}\infty - \tan^{-1}1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

18. $\sec^{-1}\left(\frac{1}{1-2x^2}\right) + 4\cos^{-1}\sqrt{\frac{1+x}{2}}$

$= \cos^{-1}\left(1-2x^2\right) + 4\cos^{-1}\sqrt{\frac{1+x}{2}}$

$\therefore \pi - \cos^{-1}\left(2x^2-1\right) + 4\cos^{-1}\sqrt{\frac{1+x}{2}}$ -- (1)

put $x = \cos\theta$ then (1) can be written as

$\pi - \cos^{-1}\left(\cos 2\theta\right) + 4\cos^{-1}\cos\left(\frac{\theta}{2}\right)$

$= \pi - 2\theta + 2\theta = \pi$ which is constant for all x.

19. Since, $\tan^{-1} x \text{ and } \cot^{-1} x$ exists for all $x \in R$ and $\cos^{-1}\left(2-x\right)$ exists, if $-1 \le 2-x \le 1$

$\therefore \tan^{-1} x - \cot^{-1} x = \cos^{-1}\left(2-x\right)$ is possible only, if $1 \le x \le 3$, thus the solution of given equation is [1,3] but at x = 3 given equation not satisifed therefore x $\in [1,3)$.

20. $\cos^{-1} x = \frac{\pi}{3}x + \frac{\pi}{6}$ has only one real root.

21. $\sqrt{1+\cos 2x} = \sqrt{2}\sin^{-1}\left(\sin x\right)$

$\left|\cos x\right| = \sin^{-1}\left(\sin x\right)$ The graph of

$y = \left|\cos x\right|$ intersect the curve

$y = \sin^{-1}\left(\sin x\right)$ in 2 points in $\left[-\pi, \pi\right]$

22. $x^2 + x + 1 = \lambda x + 1$

$\Rightarrow x = 0, \lambda - 1 \text{ and } -1 \le x \le 0$

$\Rightarrow 0 \le \lambda < 1 \Rightarrow \lambda = 0$

23. from given equation.

$\sum x_1 = \sin 2\beta; \sum x_1x_2 = \cos 2\beta; \sum x_1x_2x_3 = \cos\beta;$

and $x_1x_2x_3x_4 = -\sin\beta$

$\therefore \tan^{-1}\left(x_1\right) + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4 =$

$= \tan^{-1}\left[\frac{\sum x_1 - \sum x_1x_2x_3}{1 - \sum x_1x_2 + x_1x_2x_3x_4}\right] = \frac{\pi}{2} - \beta$

24. $\cot^{-1}\left(\frac{n}{\pi}\right) > \frac{\pi}{6} \Rightarrow \frac{n}{\pi} < \sqrt{3}$ $n < \pi\sqrt{3}$

$n < 5.46$

$\therefore$ Maximum value of 'n' is 5.

25. $\left(\cot^{-1} x\right)^2 - 7\cot^{-1} x + 10 > 0$

$\Rightarrow \left(\cot^{-1} x - 2\right)\left(\cot^{-1} x - 5\right) > 0$

Since $Cot^{-1}x - 5$ is negative and does not exits

$\therefore \cot^{-1} x < 2$ $\Rightarrow x \in \left(\cot 2, \infty\right)$

26. $\cos^{-1}\left(\cos 4\right) = \cos^{-1}\left\{\cos\left(2\pi - 4\right)\right\} = 2\pi - 4$

$\Rightarrow 2\pi - 4 > 3x^2 - 4x$

$\Rightarrow 3x^2 - 4x - \left(2\pi - 4\right) < 0$

$\Rightarrow \frac{2-\sqrt{6\pi-8}}{3} < x < \frac{2+\sqrt{6\pi-8}}{3}$

27. $\sin^{-1}(\sin 12) = \sin^{-1}\sin(12 - 4\pi) = 12 - 4\pi$

$\cos^{-1}(\cos 12) = \cos^{-1}\cos(4\pi - 12) = 4\pi - 12$

\ $(k-2)x^2 + 8x + k + 4 > 0$

if $k = 2$ then $8x + 4 > 0$ (not possible)

if $k > 2$ then $k - 2 > 0$

and $64 - 4(k-2)(k+4) < 0$

$16 < k^2 + 2k - 8$, $k^2 + 2k - 24 > 0$

$(k+6)(k-4) > 0 \Rightarrow k = 5$

28. $\cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} - \sin^{-1}x = \cos^{-1}x$

$\Rightarrow \cos\left(\cos^{-1}(2xy)\right) = \cos\left(\cos^{-1}x - \cos^{-1}y\right)$

$\Rightarrow 2xy = xy + \sqrt{1-x^2}\sqrt{1-y^2} \Rightarrow x^2 + y^2 = 1$

29. Let $f(x) = x^3 + bx^2 + cx + 1$

$f(0) = 1 > 0, f(-1) = b - c < 0$

$\Rightarrow \alpha$ lies between -1 and 0

$\Rightarrow \alpha < 0 \Rightarrow \tan(\frac{1}{\alpha}) = -\pi + \cot^{-1}\alpha$

$$\tan^{-1}\alpha+\tan^{-1}\frac{1}{\alpha}=-\pi+\tan^{-1}\alpha+\cot^{-1}\alpha$$

$$=-\pi+\frac{\pi}{2}\quad=-\frac{\pi}{2}$$

30. $$\frac{\alpha^3}{2}\operatorname{cosec}^2\left(\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right)+\frac{\beta^3}{2}\sec^2\left(\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right)$$

$$=\alpha^3\left(\frac{1}{1-\cos\left(\tan^{-1}\frac{\alpha}{\beta}\right)}\right)+\beta^3\left(\frac{1}{1+\cos\left(\tan^{-1}\frac{\beta}{\alpha}\right)}\right)$$

$$=\alpha^3\left(\frac{1}{1-\frac{\beta}{\sqrt{\alpha^2+\beta^2}}}\right)+\beta^3\left(\frac{1}{1+\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}}\right)$$

$$=\sqrt{\alpha^2+\beta^2}\left\{\frac{\alpha^3}{\sqrt{\alpha^2+\beta^2}-\beta}+\frac{\beta^3}{\sqrt{\alpha^2+\beta^2}+\alpha}\right\}$$

$$=\sqrt{\alpha^2+\beta^2}[(\alpha+\beta)\sqrt{\alpha^2+\beta^2}]$$

$$=(\alpha+\beta)(\alpha^2+\beta^2)$$

31. $x=n\pi-\tan^{-1}3$

$\Rightarrow\tan^{-1}3=n\pi-x\quad\Rightarrow\tan(n\pi-x)=3$

$\Rightarrow-\tan x=3\Rightarrow\tan x=-3$

$\Rightarrow\tan 2x=\dfrac{2\tan x}{1-\tan^2 x}=\dfrac{3}{4}$ and

$\cos x=\pm\dfrac{1}{\sqrt{1+\tan^2 x}}=\pm\dfrac{1}{\sqrt{10}}$

On substituting these value in the given equation we find only $\cos x=-\dfrac{1}{\sqrt{10}}$ satisfies the equation. So that the given equation holds for values of x for which $\tan x=-3$ and $\cos x=-\dfrac{1}{\sqrt{10}}$

Which is possible if x lies in the second quadrand only and so n must be odd integer.

LEVEL - IV

1. Assertion (A) : If $0<x<\dfrac{\pi}{2}$ **then** $\sin^{-1}(\cos x)+\cos^{-1}(\sin x)=\pi-2x$

Reason (R) : $\cos^{-1}x=\dfrac{\pi}{2}-\sin^{-1}x\ \forall\ x\in[0,1]$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

2. **Statement -1: Number of roots of the equation** $\cot^{-1}x+\cos^{-1}2x+\pi=0$ **is zero.**

Statement - 2: Ragne of $\cot^{-1}x$ **and** $\cos^{-1}x$ **is** $(0,\pi)$ **and** $[0,\pi]$ **respectively**

1) Both the statements are true and statement 2 is the correct explanation of statement 1
2) Both the statements are true but statement 2 is not the correct explanation of statement 1
3) statement 1 is true and statement 2 is false
4) statement 1 is false and statement 2 is true

3. **Let** (x,y) **be such that**

$\sin^{-1}(ax)+\cos^{-1}(y)+\cos^{-1}(bxy)=\pi/2$

a) If $a=1,b=0,then$ (x,y) lies on — p) $x^2+y^2=1$

b) If $a=1,b=1,then$ — q) $(x^2-1)(y^2-1)=0$

c) $a=1,b=2,then$ — r) $y=x$

d) $a=2,b=2then$ — s) $(4x^2-1)(y^2-1)$

1) $a\to p,b\to q,c\to p,d\to s$
2) $a\to p,b\to q,c\to s,d\to r$
3) $a\to r,b\to q,c\to s,d\to r$
4) $a\to p,b\to q,c\to r,d\to s$

4. **Which of the following function(s) is/are identical**

(i) $f(x)=\cot(\cot^{-1}x)$; $g(x)=x$

(ii) $f(x)=\sin^{-1}(3x-4x^3)$; $g(x)=3\sin^{-1}x$

(iii) $f(x) = \sec^{-1}x + \mathrm{cosec}^{-1}x$; $g(x) = \frac{\pi}{2}$

(iv) $f(x) = \tan(\cot^{-1}x)$; $g(x) = \cot(\tan^{-1}x)$

(v) $f(x) = e^{\ln \cot^{-1} x}$; $g(x) = \cot^{-1}x$

(vi) $f(x) = e^{\ln \sec^{-1} x}$; $g(x) = \sec^{-1}x$

1) only i,iv,v 2) i, iii,iv only

3) ii,v,vi only 4) i,iv only

5. **Which of the following function(s) is/are periodic?**

(A) $f(x) = \frac{2^x}{2^{[x]}}$ **where [ ] denotes greatest integer function**

(B) $g(x) = \mathrm{sgn}\ \{x\}$ **where** $\{x\}$ **denotes the fractional part function**

(C) $h(x) = Sin^{-1}\left(\cos(x^2)\right)$

(D) $k(x) = Cos^{-1}\left(\sqrt{\sin x}\right)$

1) only A,B,D 2) only B,D,C

3) only B,A,C 4) only A,D,C

6. **Solution of the system of inequalities involving inverse circular functions** $\operatorname{arc}\tan^2 x - 3\operatorname{arc}\tan x + 2 > 0$ **and** $[\sin^{-1}x] > [\cos^{-1}x]$ **where [ ] denotes the greatest integer function.**

1) $x \in (\sin 1, 1)$ 2) $x \in [\sin 1, 1]$

3) $x \in [\sin 1, 1)$ 4) $x \in (\sin 1, 1]$

LEVEL - IV-KEY

01) 1	02)1	03) 1	04) 1
05) 1	06) 2		

LEVEL - IV-HINTS

1) A : $\sin^{-1}\left(\sin\left(\frac{\pi}{2}-x\right)\right)+\cos^{-1}\left(\cos\left(\frac{\pi}{2}-x\right)\right)$

Reason : $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ is correct.

3) $\sin^{-1}(ax)+\cos^{-1}y+\cos^{-1}(bxy)=\frac{\pi}{2}$

$\sin^{-1}(ax)+\cos^{-1}(bxy)=\frac{\pi}{2}-\cos^{-1}y$

use $\sin^{-1}x+\sin^{-1}y$ for LHS

we get $abx^2y+\sqrt{1-b^2x^2y^2}\sqrt{1-a^2x^2}=y$

$\Rightarrow abx^2y-y=-\sqrt{1-b^2x^2y^2}\sqrt{1-a^2x^2}$

squaring on both sides and simplify we get

$y^2+a^2x^2+b^2x^2y^2-2abx^2y^2=1$

Now verify the options

5) A) period of $x-[x]$ is 1

B) period of $\{x\}$ is 1

C) period does not exist.

D) period is 2π

6) $\left(\tan^{-1}x\right)^2-3\tan^{-1}x+2>0$ or

$t^2-3t+2>0$, $(t-2)(t-1)>0$

$\left(\tan^{-1}x-2\right)$ is always <0

$\tan^{-1}x<1 \Rightarrow x<\tan 1$, $\therefore x\in(-\infty,\tan 1)$

again $\left[\sin^{-1}x\right]>\left[\cos^{-1}x\right]$

$\left[\sin^{-1}x\right]$ can take the values $\{-2,-1,0,1\}$

and $\left[\cos^{-1}x\right]$ can take the values $\{0,1,2,3\}$

$\therefore\left[\sin^{-1}x\right]$ can be greater than $\cos^{-1}x$ only.

If $\left[\sin^{-1}x\right]=1$ and $\left[\cos^{-1}x\right]=0$

Now $\left[\sin^{-1}x\right]=1$, $1\le\sin^{-1}x<2$

$\therefore \sin 1\le x\le 1$ and $\left[\cos^{-1}x\right]=0$

$0\le\cos^{-1}x<1$, $\cos 1\le x\le 1$

$\therefore$ Now x must satisfy $x\in[\sin 1, 1]$